THREE ARITHMETIC INPUTS PROVED AND ONE REMAINING: A VERIFIED MATHEMATICAL ACCOUNT

Abstract. We verify the proposed removal of arithmetic hypotheses from the delivered manuscript on a Diophantine definition of the ordinary integers in the rational numbers. Three original statements are established in their full required scope: the rank-one Mordell–Weil assertion E, the five-point odd-contact height estimate H, and the simultaneous compatible prime-pattern assertion R. For H the exponent can be taken to be 24. We include their proofs and the complete conditional passage from the remaining pole-parity statement P to both an integer model and its scalar evaluation graph. The original theorem therefore follows from P alone, with every rational scalar, signed index, and exceptional marked point retained. Conversely, the scalar conclusion implies the existence assertion P. This equivalence proves neither side. We also prove exact reductions of the proposed parity construction and identify its unresolved rational-point step. Finite two-descent permits a divisible 2-primary Tate–Shafarevich alternative; no claim of unconditional integer definability is made. The companion converse theorem is not a premise of the results established here.

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1. The exact target and the verified outcome

A subset of $\mathbb{Q}^r$ is *Diophantine over* $\mathbb{Q}$ if it is the projection of the rational zero set of one polynomial with rational coefficients. The question here concerns the ordinary subset $\mathbb{Z} \subset \mathbb{Q}$. A Diophantine model of the integer ring on elliptic points does not by itself define that subset: one must also identify a rational scalar with a point's integer index. Both parts of the delivered theorem are retained below.

1.1. Documents and scope.

The target is Theorem 1.1 of `paper_rewritten.tex`, dated 27 October 2026, with its 19-page PDF [1]. Its source SHA-256 is

bc5fb24c479693e9180d220e9ce4f2c0707eb65f9e49fc4e38a8c5c2946fbf97.

The proposed solution being checked is the 30 October 2026 `research_report.tex`, with its 28-page PDF [2]; its source SHA-256 is

738666e2f2a4e541333049810598c3e31c25bd4187fa0530f209ff82832b1f59.

The preceding diagnostic brief [3] supplies context, not a substitute for either actual source. The September source [4] is used at the pinned repository version recorded in the bibliography. In particular, its oracle-reduction conclusion is not substituted for the stronger scalar definition asserted by the current manuscript.

The fixed arithmetic setting is

$$F = \mathbb{Q}(\sqrt{2}), \qquad d = 75 - 53\sqrt{2}, \qquad E : y^2 = x^3 - d^2 x.$$

All quantifiers in the formulas ultimately range over rational numbers; points in $E(F)$ are finite rational tuples. No genericity, positivity, or bounded-height restriction is imposed on the integer indices or rational scalars in the target. The rank, height, and prime-pattern assertions are proved below and are not moved into unstated standing hypotheses.

Theorem 1.1 (Unconditional arithmetic inputs)**.** *Inputs E, H, and R, reproduced in Section 2, hold. Their proofs use neither P nor the desired Diophantine definition. In H one may take $c_0 = 24$ for the fixed rational five-point configuration constructed in Section 4. R holds for every fixed configuration of the stated kind, and hence for that same configuration.*

Theorem 1.2 (The original conclusion under P alone)**.** *Assume Input P of Section 2. Fix any integer $m_0 \geq 1$ and nontorsion point $P \in E(F)$ such that $G = m_0 E(F) = \mathbb{Z}P$; such a pair exists by Theorem 1.1. Then:*

(i) *There is a Diophantine relation* Mul_G *on the rational codes of G^3 such that*

$$\mathrm{Mul}_G(n_1P, n_2P, n_3P) \iff n_1 n_2 = n_3 \qquad (n_1, n_2, n_3 \in \mathbb{Z}).$$

Together with point addition, it makes $n \mapsto nP$ a Diophantine model of $(\mathbb{Z}, +, \cdot)$ over $\mathbb{Q}$.

(ii) *There is a Diophantine relation* $\mathrm{Eval} \subseteq \mathbb{Q} \times G$ *satisfying*

$$\mathrm{Eval}(x, nP) \iff x = n \qquad (x \in \mathbb{Q},\ n \in \mathbb{Z}).$$

Consequently there are an integer $N \geq 1$ and a polynomial $D \in \mathbb{Z}[X, Y_1, \ldots, Y_N]$ such that

$$x \in \mathbb{Z} \iff \exists \mathbf{y} \in \mathbb{Q}^N \ D(x, \mathbf{y}) = 0 \qquad (x \in \mathbb{Q}).$$

Denote the last scalar existence statement by $T_{\mathbb{Z}}$. The complete argument in Sections 6–8 proves the theorem; it does not appeal to the preceding report's assertion that the argument works. Those sections assume only P, with E, H, and R supplied by their independent proofs. In Section 9 we obtain the unconditional equivalence

$$\mathrm{P} \iff T_{\mathbb{Z}}.$$

This is an equivalence between existence assertions, not a proof of either assertion or of the source's particular proposed formula.

Outcome. The result is a rigorously improved conditional theorem: E, H, and R have been eliminated as assumptions; P remains unresolved. No counterexample to P or to the original theorem is established. The original conditional argument survives the audit, including all signed and zero indices and the finite exceptional set in the evaluation formula.

Input	Final status	Exact role and scope
P	Unproved	One fixed existential formula accepts every integer and excludes every negative odd valuation outside one fixed finite set. Used in the square guards and local certificates.
E	Proved in full	The specified curve has rank one. Killing torsion gives a fixed infinite cyclic image; no assertion that the displayed point is a primitive generator is required.
H	Proved in full	One fixed five-point configuration, every rational parameter outside it, and fixed constants with exponent 24. Controls false multiplication witnesses and scalar heights.
R	Proved in full	One finite multiplier catalogue before A and the prime lower bound; every signed A and arbitrarily large simultaneous compatible primes. Supplies completeness of the comparisons.

1.2. **Why the proof is grounded.** The rank proof is a two-isogeny descent. The height proof uses a fixed quaternionic Shimura curve and verified effective Shafarevich estimates; it requires neither a parity formula nor an integer model. The prime-pattern proof uses finite-complexity prime theory and explicit sieve bounds, with the coefficient and cutoff dependencies kept separate.

After these inputs are supplied, a false multiplication tuple has nonzero integer discrepancy $\delta = n_1 n_2 - n_3$. Local certificates force every relevant odd-contact prime to divide δ to a fixed power. H bounds coordinate heights in terms of $|\delta|$. Testing coordinates of one fixed multiple of the points contradicts canonical-height growth. This first constructs the multiplication relation. Only then is the Davis–Putnam–Robinson–Matiyasevich theorem used to express computable relations on the point indices. A support relation and a 3-adic congruence of sufficient, internally represented precision force scalar equality.

The indispensable local estimates, point encodings, and fixed-local-ring formula are proved in the appendices. The resulting dependency order is

$$\begin{gathered}\text{independent proofs of E, H, R,}\\ \mathrm{P} \implies \text{local certificates} \implies \mathrm{Mul}_G \implies \text{integer relations on } G,\\ \text{support and congruence} \implies \mathrm{Eval} \implies T_{\mathbb{Z}}.\end{gathered}$$

The reverse implication $T_{\mathbb{Z}} \Rightarrow$ P is immediate from the scalar definition and is used only to describe the residual problem, never to prove P.

2. The original inputs and fixed conventions

The four statements below retain their original labels and formulations. Their reproduction does not assume E, H, or R: Sections 3–5 prove them. When P is assumed, fix its formula and exceptional set once and for all. All subsequent data in this section are fixed. A formula is *positive existential* if it is built from polynomial equations by finite conjunctions, finite disjunctions, and existential quantifiers. All scalar variables in such a formula range over $\mathbb{Q}$. Over $\mathbb{Q}$, positive existential definitions are equivalent to Diophantine definitions: sums of squares combine equations, products combine alternatives, and $a \neq 0$ is written $\exists b\ (ab = 1)$. We may therefore use a finite collection of named predicates, provided each has an explicit positive existential definition.

We use $\mathbb{N} = \{0, 1, 2, \ldots\}$ and the convention $v_q(0) = +\infty$. For a rational prime q, $v_q(q) = 1$. Heights are absolute logarithmic Weil heights, with natural logarithms; in particular,

$$h(u/v) = \log \max\{|u|, v\} \quad \text{if } u, v \in \mathbb{Z},\ v > 0,\ \gcd(u, v) = 1.$$

We use the familiar inequalities $h(ab) \leq h(a) + h(b)$ and $h(a + b) \leq h(a) + h(b) + \log 2$, also over number fields.

2.1. **The four inputs.**
Input P: parity of poles. There are a positive existential formula $\Phi(t)$ over $\mathbb{Q}$ and a finite set S_Φ of rational primes such that

$$\Phi(t) \text{ holds for every } t \in \mathbb{Z}, \qquad \Phi(t),\ q \notin S_\Phi,\ v_q(t) < 0 \implies v_q(t) \in 2\mathbb{Z}. \tag{2.1}$$

Thus Φ excludes odd-order poles outside S_Φ. It is not assumed to define integrality.
Input E: a cyclic subgroup of a Mordell–Weil group. Set

$$F = \mathbb{Q}(\sqrt{2}), \qquad d = 75 - 53\sqrt{2}, \qquad E:\ y^2 = x^3 - d^2x.$$

Write σ for the nontrivial automorphism of $F/\mathbb{Q}$. There are a fixed integer $m_0 \geq 1$ and a fixed nontorsion point $P \in E(F)$ such that

$$G := m_0E(F) = \mathbb{Z}P. \tag{2.2}$$

Here $E(F)$ is the Mordell–Weil group of rational points over F, and G is its chosen infinite cyclic subgroup of finite index. The equality concerns rational points, rather than the algebraic image of an isogeny. The *index* of $T \in G$ is the unique $n \in \mathbb{Z}$ with $T = nP$.
Input H: a height estimate from five marked points. There is a set $\mathcal{C} = \{b_0, \dots, b_4\}$ of five distinct points of $\mathbb{P}^1(\mathbb{Q})$, with $b_0 = \infty$, and fixed primitive integral linear forms

$$L_0(U, V) = V, \qquad L_i(U, V) = \alpha_iU + \beta_iV \quad (1 \leq i \leq 4)$$

vanishing at the corresponding points. Thus $\alpha_i \neq 0$ for $i \geq 1$. There are a finite prime set S_0 and constants $H_0, c_0 > 0$ such that

$$h(s) \leq H_0M_{S_0}(s)^{c_0} \qquad (s \in \mathbb{P}^1(\mathbb{Q}) \setminus \mathcal{C}).$$

For $s = (u : v) \in \mathbb{P}^1(\mathbb{Q}) \setminus \mathcal{C}$ and a prime q, its *normalized contact order* with b_i is

$$v_q(L_i(u, v)) - \min\{v_q(u), v_q(v)\}. \tag{2.3}$$

This is independent of the chosen homogeneous coordinates. The number $M_{S_0}(s)$ is the product of the primes $q \notin S_0$ for which at least one of these five orders is odd. With primitive integral coordinates, the orders are simply the valuations of the five nonzero form values. Thus this product records odd orders, rather than all positive contact orders.
Input R: compatible prime-pattern representations. There is a fixed finite set $\mathcal{K} \subset \mathbb{Z}_{>0}$ with the following property. For every integer A and every prescribed lower bound, there are integers u_1, u_2, u_3, v, with $v \neq 0$, such that

$$Av = u_1 + u_2 + u_3, \tag{2.4}$$

and each of the thirteen entries

$$v, \qquad L_i(u_j, v) \quad (1 \leq i \leq 4,\ 1 \leq j \leq 3) \tag{2.5}$$

is $\epsilon\kappa r$, with $\epsilon \in \{1, -1\}$, $\kappa \in \mathcal{K}$, and r a positive prime exceeding that bound. Every selected prime splits in F and is a prime of good reduction for both displayed curves E and E^σ. At either place above r, if d_1 and d_2 are the orders of their reduction groups, then

$$\gcd(d_1, d_2) \mid 4, \qquad r \nmid d_1d_2. \tag{2.6}$$

The set $\mathcal{K}$ is independent of both A and the lower bound. There is no primality predicate in our formulas: primes enter the proof that integer inputs have witnesses.

These four inputs correspond, respectively, to the source labels `par:pole-parity`, `ell:rank`, `hei:height`, and `pri:pattern` in the dated manuscript [4]. The constructions below require their compatible data, and not merely an undecidability assertion.

2.2. **One exceptional set and one height constant.** Use Theorem 4.1 and set $c = c_0 = 24$. Choose a finite prime set S containing S_0, S_Φ, 2, 3, the primes ramified in F, all bad primes for the two displayed integral equations, every prime at most $c+1$, and the prime divisors of

$$\prod_{i=1}^{4} \alpha_i, \qquad \prod_{0\le i<j\le 4} \det(L_i, L_j), \qquad \prod_{\kappa\in\mathcal{K}} \kappa.$$

The determinants are nonzero because the marked points are distinct. Enlarging the exceptional set discards at most one factor q for each newly excluded prime. We may therefore fix an integer $H \ge 1$ such that

$$h(s) \le H M_S(s)^c \qquad (s \in \mathbb{P}^1(\mathbb{Q}) \setminus \mathcal{C}). \tag{2.7}$$

For instance any integer at least $H_0(\prod_{q\in S\setminus S_0} q)^c$ suffices after enlarging it to be at least 1. Neither S nor H will vary with an integer index or a requested congruence precision.

The canonical-height comparison and the nontorsion of P give a fixed integer $a_* \ge 1$ such that

$$h(x(tP)) \ge \frac{t^2}{a_*} - a_* \qquad (t \in \mathbb{Z} \setminus \{0\}). \tag{2.8}$$

Indeed, the canonical height of tP is $t^2\widehat{h}(P)$, with $\widehat{h}(P) > 0$, and its difference from a fixed positive multiple of $h(x(tP))$ is bounded [5, Chapter VIII]; see also the height calculation in the proof of [6, Lemma 8].

2.3. **Point notation inside rational formulas.** We encode $u_0 + u_1\sqrt{2} \in F$ by $(u_0, u_1) \in \mathbb{Q}^2$. A point has an identity flag and affine coordinates, giving a unique rational code for every point of $E(F)$. Membership in G is the existential condition $T = [m_0]U$ for some $U \in E(F)$. Appendix B gives equations for these codes and the group operations. Thus a quantified point in G always means a finite rational tuple with these conditions imposed. Fixed integer multiples and conjugation are allowed in this notation. Multiplication by a *variable* integer is not allowed until Mul_G has been constructed.

On nonidentity points of G put $z = -x/y$, and set $z(O) = 0$. A point of the displayed curve with $x = 0$ or $y = 0$ is a point of order two. Since G is torsion-free, x, y, z are nonzero on $G \setminus \{O\}$. The same applies to the conjugate group. For a finite place w write F_w for the completion, $\mathcal{O}_{F_w}$ for its valuation ring, and $\mathcal{O}_{F,w} = F \cap \mathcal{O}_{F_w}$ for the corresponding localization of $\mathcal{O}_F$. At good reduction a point is *formal at* w if it lies in the kernel of reduction; the parameter z then belongs to the maximal ideal. On a nonidentity point this is equivalent to $w(1/x) > 0$.

Let ℓ_k be the degree-k truncation of the formal logarithm in the parameter z, for an integer $k \ge 1$. Its coefficients are computed from the fixed equation. Only ℓ_c, a fixed polynomial, will occur in the multiplication formula. Infinite series are used in proofs of estimates, not as conditions in an existential formula. We also use the rational projection

$$\rho : F \longrightarrow \mathbb{Q}, \qquad \rho(u_0 + u_1\sqrt{2}) = u_0 = \tfrac{1}{2}\,\mathrm{Tr}_{F/\mathbb{Q}}(u_0 + u_1\sqrt{2}).$$

All constants and all polynomials used below are fixed finite data. The results assert the existence of the resulting formulas and algorithms; they do not supply numerical values for every imported constant or an expanded coefficient list for the final polynomial.

3. Input E is unconditional

The following verifies the descent given in the retrieved elliptic source [4]; the local and index calculations are included so that no numerical rank report is being used.

Proposition 3.1. *For the specified curve,* $\mathrm{rank}\, E(F) = 1$. *Consequently there exist* $m_0 \ge 1$ *and a nontorsion point* P *with* $m_0 E(F) = \mathbb{Z}P$.

Proof. The ring $\mathcal{O}_F = \mathbb{Z}[\sqrt{2}]$ has class number one: its Minkowski bound is $\sqrt{8}/2 = \sqrt{2} < 2$. Its unit squareclass group has order four, and -1 and $1+\sqrt{2}$ realize its four real signature patterns. A totally positive unit is therefore a square. The element d is totally positive of norm 7, so $\mathfrak{d} = (d)$ is prime with residue field $\mathbb{F}_7$.

Use the two-isogenous curve

$$E' : y^2 = x^3 + 4d^2x$$

and squareclass homomorphisms $\alpha : E(F) \to F^*/F^{*2}$ and $\alpha' : E'(F) \to F^*/F^{*2}$. Away from O and $(0,0)$ these send a point to its x-class; the exceptional values are 1 and the respective linear coefficient. Their kernels are the images of the dual two-isogenies. These are the usual two-isogeny descent maps [5, Chapter X]; they can also be checked from the maps recorded after this proof.

For $y^2 = x^3 + bx$ at a finite place where b is a unit, a nonzero x has even valuation. Indeed for negative valuation the dominating term has valuation $3v(x)$, and for positive valuation it has valuation $v(x)$; neither can be odd. Thus $\operatorname{im}\alpha$ is supported only at $\mathfrak{d}$, up to unit squares, and has at most eight elements. The four two-torsion points give the distinct classes $1, -1, d, -d$. Put $w = -5 + 4\sqrt{2}$. Direct multiplication gives

$$w = d(1+\sqrt{2})^3, \qquad w^3 - w = -8d,$$

so

$$Q = (-dw, 2\sqrt{2}\,d^2) = (799 - 565\sqrt{2}, -31800 + 22486\sqrt{2}) \in E(F).$$

Its x-coordinate has opposite real signs, unlike the four preceding classes. Hence $|\operatorname{im}\alpha| = 8$.

For E', every nonzero x is totally positive. All its finite valuations are even away from the primes over 2 and 7 by the preceding observation. At the prime over 2, normalized to have integral value group, $v(4d^2) = 4$. If $e = v(x)$ were odd, the valuations $3e$ and $4 + e$ would be distinct and their minimum odd, which is impossible. At the conjugate prime above 7, $4d^2$ is a unit, so the first unit-coefficient argument already applies. At $\mathfrak{d}$ the same reasoning, with coefficient valuation 2, leaves only $e = 1$ to check. Writing $x = du$ with u a unit would give

$$y^2 = d^3u(u^2 + 4).$$

Even valuation would require $u^2 = -4$ in $\mathbb{F}_7$, but $-4 \equiv 3$ is not a square there. Thus every finite valuation of x is even. Class number one makes x a square times a unit; total positivity makes that unit a square. The identity and $(0,0)$ also have trivial class, so $|\operatorname{im}\alpha'| = 1$.

The kernel of the dual isogeny is contained in the first isogeny's image, because $(\pm d, 0)$ map to $(0,0) \in E'(F)$. The index formula is therefore

$$|\operatorname{im}\alpha|\,|\operatorname{im}\alpha'| = [E(F) : 2E(F)] = 2^{r+2},$$

where $r = \operatorname{rank} E(F)$ and the factor 4 comes from the full rational two-torsion. Thus $8 = 2^{r+2}$, giving $r = 1$. Mordell–Weil now allows m_0 to be any multiple of the exponent of the finite torsion subgroup; its image is torsion-free of rank one, hence has a generator P. □

For an explicit check of the descent interface, the two maps on their finite nonkernel charts are

$$\varphi(x,y) = \left(x - \frac{d^2}{x},\ y\left(1 + \frac{d^2}{x^2}\right)\right),$$

$$\widehat{\varphi}(X,Y) = \left(\frac{X + 4d^2/X}{4},\ \frac{Y(1 - 4d^2/X^2)}{8}\right).$$

They compose to [2]. Their coordinate formulas show that their images have trivial squareclass on the target. Conversely, if $X = u^2 \neq 0$ on E', then $t = (X + Y/u)/2$ gives a preimage (t, ut) on E; if $x = v^2 \neq 0$ on E, then $(X,Y) = (2x + 2y/v, 2vX)$ gives a preimage on E'. The exceptional points have the stated classes. This also verifies the kernel and index calculation without a saturation computation.

No assertion that the displayed Q is a primitive generator is needed or made. The explicit point together with the descent proves rank one, not a numerical value for its index in the free part.

4. A proof of the five-point height estimate

The height input can be supplied without any of the other three inputs. We retain the quaternionic configuration underlying the source's construction, but use an existing effective Shafarevich theorem for the arithmetic estimate. Throughout this section, h denotes absolute logarithmic Weil height and h_F stable Faltings height. Fixed dimensional differences between conventions for the latter are absorbed into constants.

Theorem 4.1. *There exist five distinct points $\mathcal{C} = \{b_0, \dots, b_4\} \subset \mathbb{P}^1(\mathbb{Q})$, with $b_0 = \infty$, primitive integral forms*

$$L_0(U,V) = V, \qquad L_i(U,V) = \alpha_i U + \beta_i V \quad (1 \le i \le 4), \qquad \alpha_i \neq 0,$$

vanishing at the corresponding points, a finite set S_0 of rational primes, and a constant $H_0 > 0$ such that

$$h(s) \le H_0 M_{S_0}(s)^{24} \qquad (s \in \mathbb{P}^1(\mathbb{Q}) \setminus \mathcal{C}).$$

Here $M_{S_0}(s)$ is the product of the primes $q \notin S_0$ for which at least one normalized contact order

$$v_q(L_i(u,v)) - \min\{v_q(u), v_q(v)\}, \qquad s = (u:v),$$

is odd. In particular, Input H holds with $c_0 = 24$.

The proof has two parts. A fixed cover of the projective line carries abelian surfaces and ramifies to order two exactly at five rational points. Even contact therefore introduces no new ramification in a specialized fiber field. The surfaces have quaternionic multiplication and are isogenous to their Galois conjugates. These properties put them within the scope of von Känel–Kret's effective height bounds. A comparison with the height on the base then proves the theorem.

4.1. **The rational five-point configuration.** Let $B/\mathbb{Q}$ be the indefinite division quaternion algebra of discriminant $210 = 2 \cdot 3 \cdot 5 \cdot 7$, and fix a maximal order $\mathcal{O}$. Consider the maximal-level Shimura curve

$$X = \mathrm{Sh}_{\widehat{\mathcal{O}}^\times}(B^\times, \mathfrak{H}^\pm).$$

Its reflex field is $\mathbb{Q}$: after splitting B, the weight-one cocharacter has the usual GL_2 conjugacy class, which is Galois invariant. Canonical models, level maps, and finite-adelic right actions consequently exist over $\mathbb{Q}$; see [7, Theorem 13.6, Remark 13.8, and Section 14] and [8]. The curve is proper because B is a division algebra.

Strong approximation for B^1, together with the reduced-norm theorem and $\mathrm{Nrd}(\mathcal{O}_p^\times) = \mathbb{Z}_p^\times$, gives

$$B^\times(\mathbf{A}_f) = B^\times(\mathbb{Q})^+ \widehat{\mathcal{O}}^\times.$$

Indeed, one first matches the finite-adelic norm modulo $\widehat{\mathbb{Z}}^\times$ by a positive rational reduced norm, and then uses strong approximation on the norm-one factor. Negative rational reduced norms also exist, since there is no ramified real place. Thus $X_{\mathbf{C}}$ is connected and

$$X(\mathbf{C}) = (\mathcal{O}^1/\{\pm 1\})\backslash\mathfrak{H}.$$

The strong-approximation and type-number-one statements used here are [9, Theorems 28.2.10–28.2.11].

There are no elliptic stabilizers in $\mathcal{O}^1/\{\pm 1\}$. A noncentral integral norm-one element fixing a point has integral trace of absolute value less than two, so generates $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-3})$. The former splits at the ramified prime 5 and the latter at the ramified prime 7; neither embeds in B. The area formula [9, Theorem 39.1.2] now gives

$$\frac{\mathrm{area}(X(\mathbf{C}))}{2\pi} = \frac{1}{6} \prod_{p \mid 210} (p-1) = 8.$$

There are no cusps or elliptic corrections, and hence $g(X) = 5$.

The finite-adelic normalizer of $\widehat{\mathcal{O}}$ induces over $\mathbb{Q}$ the Atkin–Lehner group

$$\mathcal{W} \simeq (\mathbb{Z}/2\mathbb{Z})^4.$$

At a split finite place the normalizer is scalars times order units; at a division place its quotient by scalars and order units has order two, detected by parity of norm valuation. Adelic scalars act trivially because $\mathbf{A}_f^\times = \mathbb{Q}^\times \widehat{\mathbb{Z}}^\times$. The resulting action is faithful: a generic half-plane point has rational stabilizer equal to the center, accounting exactly for this scalar kernel. Put $X^* = X/\mathcal{W}$.

Lemma 4.2. *The quotient X^* is isomorphic over $\mathbb{Q}$ to $\mathbb{P}^1$. The map $X \to X^*$ has exactly five branch values, each individually rational, and every ramification index is two.*

Proof. A finite stabilizer acting on a characteristic-zero smooth curve acts faithfully on the tangent line and is cyclic. Since $\mathcal{W}$ has exponent two, each nontrivial stabilizer has order two. If r is the number of geometric branch values, Riemann–Hurwitz gives

$$2g(X^*) - 2 + \frac{r}{2} = \frac{2g(X) - 2}{16} = \frac{1}{2}, \qquad r = 5 - 4g(X^*) \le 5. \tag{4.1}$$

For each $m \in \{30, 42, 70, 105, 210\}$, the quadratic field $\mathbb{Q}(\sqrt{-m})$ is nonsplit at every prime dividing 210. For $m = 30, 42, 70$, the only remaining primes are $7, 5, 3$, respectively, and the residues of $-m$ there are the nonsquares $5, 3, 2$. For $m = 105$, the quadratic discriminant is -420, so 2 is ramified as well; for $m = 210$, all four primes ramify. The quadratic embedding criterion for quaternion algebras therefore embeds each field into B.

The embedded $\sqrt{-m}$ belongs to a maximal order. At a split local place an integral element preserves a lattice; at a division place use the unique maximal order. Intersecting the local orders gives a global one containing the element. Type number one allows us to conjugate it into the fixed $\mathcal{O}$. It normalizes $\mathcal{O}$, since it is a unit away from 210 and the maximal order at a division prime is unique. Its norm is m, so it represents the involution w_m with that norm-parity label. Its square is negative, so it fixes a point in the half-plane.

The five labels give five distinct branch values. A quotient fiber is one $\mathcal{W}$-orbit, and stabilizers along that orbit are equal because $\mathcal{W}$ is abelian. Equation (4.1) forces $r = 5$ and $g(X^*) = 0$. There is consequently exactly one branch value for each label. Every w_m is defined over $\mathbb{Q}$, so Galois fixes its uniquely labelled branch value. All five values are rational. A smooth genus-zero curve with a rational point is isomorphic to $\mathbb{P}^1$ over $\mathbb{Q}$. □

Fix an isomorphism $X^* \simeq \mathbb{P}^1$ taking one branch value to infinity, call the five values $\mathcal{C}$, and choose the forms L_i in Theorem 4.1. Their numerical coordinates are not needed for the existential formulation of H. The configuration is fixed once and for all and does not depend on the point whose height will be bounded.

4.2. **An abelian surface family on an unramified level cover.**

Lemma 4.3. *There is a fixed smooth projective curve $Y/\mathbb{Q}$, possibly with several geometric components, a finite map $\pi : Y \to \mathbb{P}^1$, and a principally polarized abelian surface scheme $\mathcal{A} \to Y$ with the following properties:*

(1) The ramification index is two at every point over $\mathcal{C}$ and one elsewhere.

(2) Every geometric fiber has B-multiplication. Fibers over a common base point are geometrically isogenous.

(3) The polarized moduli map is nonconstant on every geometric component of Y.

Proof. On the rational vector space $V = B$, choose a pure quaternion Δ with $\Delta^2 < 0$ and put

$$\rho(g)(x) = x\bar{g}, \qquad \psi(x, y) = \mathrm{Trd}(\Delta x \bar{y}).$$

For example, use an embedded $\mathbb{Q}(\sqrt{-210})$ to choose Δ. The form ψ is alternating and nondegenerate, and

$$\psi(x\bar{g}, y\bar{g}) = \mathrm{Nrd}(g)\psi(x, y).$$

Conjugation and composition of right multiplications both reverse products, so ρ is a faithful group homomorphism into the symplectic-similitude group.

The standard weight-one Hodge structures of the half-plane act by right multiplication. At the real base point $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$,

$$\psi(x, x\bar{J}) = \mathrm{tr}(J\Delta x x^{\mathrm{t}}).$$

The matrix $J\Delta$ is symmetric with positive determinant and hence definite. Choose the sign of Δ so it is positive on one component. The similitude identity gives the usual polarization sign on the other component. Thus ρ is a morphism to the genus-two Siegel Shimura datum.

Choose a rational symplectic basis and the corresponding self-dual lattice. Choose a sufficiently small compact open $K_Y \subset \widehat{\mathcal{O}}^\times$ whose image lies in a principal Siegel level $N \geq 3$, and set

$$Y = \mathrm{Sh}_{K_Y}(B^\times, \mathfrak{H}^\pm).$$

Canonical-model functoriality gives the map to the fine Siegel moduli scheme over $\mathbb{Q}$. The exact representability assertion over $\mathbb{Q}$, with full symplectic-similitude level $N \geq 3$, is [8, Section 4.16 and Proposition 4.17, p. 150]; its identification with the canonical model is Theorem 4.21 there. Pullback of its universal family gives $\mathcal{A} \to Y$. The self-dual lattice need not be preserved by the whole maximal order: the intersection of levels supplies the integral family, and the rational representation retains the B-action.

The level map $Y \to X$ is unramified, because the uniformizing group of X has no elliptic stabilizers. The composite $\pi : Y \to X \to X^* \simeq \mathbb{P}^1$ therefore has the asserted ramification.

Left multiplication by B commutes with the right complex structures. The equivalence between rational weight-one Hodge structures and abelian varieties up to isogeny supplies the geometric endomorphisms; homomorphisms of abelian varieties over $\overline{\mathbb{Q}}$ do not acquire new elements after extension to $\mathbf{C}$. Changing level or acting by the finite-adelic normalizer changes the finite-adelic coordinate but preserves the rational Hodge structure, after identifying domain coordinates up to $B^\times(\mathbb{Q})$. Thus fibers with the same image in X^* are geometrically isogenous. The preceding rational-Hodge argument includes CM fibers. For comparison, [10, Lemma 1] retains an isogeny description with its quotient in that case; Corollary 2 and the opening argument of Section 3 of that paper impose the additional non-CM hypothesis. We do not use that restricted assertion for CM fibers.

Finally, complex structures vary in a half-plane, whereas a fixed rational Hodge structure has only countably many presentations on the fixed rational space B. A whole geometric component therefore cannot have a single polarized isomorphism class of fibers. Its algebraic polarized moduli map is nonconstant. □

In particular, if $s \in \mathbb{P}^1(\mathbb{Q})$ and $\pi(y) = s$, then $\pi({}^\sigma y) = s$ for every $\sigma \in G_\mathbb{Q}$. The fiber $\mathcal{A}_y$ is geometrically isogenous to every Galois conjugate.

4.3. **Comparison of base height and Faltings height.**

Lemma 4.4. *There is a constant C, depending only on the fixed family and map, such that*

$$h(\pi(y)) \leq C\big(1 + \max\{0, h_F(\mathcal{A}_y)\}\big) \qquad (y \in Y(\overline{\mathbb{Q}})).$$

Proof. Treat each geometric component after one fixed constant-field extension. Fix an even theta level, for example $r = 2$. Over a finite extension of the function field of the component, choose a symmetric ample line bundle representing the principal polarization, a rigidification, the required r^2-torsion points, and good theta data as in [11, Sections 2.3.1–2.3.4]. These choices require only a finite extension. They spread over a nonempty open of one fixed smooth projective covering curve Y'.

The algebraic theta-null construction gives a rational map $\theta : Y' \dashrightarrow \mathbb{P}^{r^4-1}$, which extends over the smooth projective curve. On that open, the height of the theta-null point is Pazuki's $h_\Theta(\mathcal{A}_y)$, up to the fixed bounded difference between the Euclidean and sup projective heights. The finite coordinate ambiguities in the construction preserve this bound.

The map is nonconstant. Indeed, [11, Section 2.3.3, p. 29] states that its theta-null point determines a polarized abelian variety up to finite ambiguity. A constant theta-null map would therefore make the polarized moduli map constant, contrary to Lemma 4.3. Consequently $\theta^*\mathcal{O}(1)$ has positive degree and is ample. If $f : Y' \to \mathbb{P}^1$ is the composite with π, choose a fixed integer a sufficiently large that

$$(\theta^*\mathcal{O}(1))^{\otimes a} \otimes (f^*\mathcal{O}(1))^{-1}$$

also has positive degree. Its height is bounded below, so the height machine gives

$$h(f(z)) \le a\, h(\theta(z)) + O(1).$$

Corollary 1.3(2) of [11], at fixed dimension and level, implies

$$h_\Theta(\mathcal{A}_y) \ll 1 + \max\{0, h_F(\mathcal{A}_y)\}.$$

This proves the required inequality on the chosen open. The omitted geometric points form a finite set, so a larger constant handles them. Absolute heights and stable Faltings heights are unchanged by finite field extension; in particular the constants do not depend on the residue-field degree of y. Take a maximum over the finitely many components. □

4.4. **Specialized fields and odd contact.** Enlarge a fixed finite prime set S, containing 2, so that the curves, map, and abelian scheme spread over $\mathbb{Z}[S^{-1}]$ with the following properties: the curves are smooth and proper, the five branch sections are disjoint, π is finite and étale away from them, and

$$\pi^* b = 2D_b$$

at each branch section, where D_b is reduced and finite étale over the ground ring. Remove also primes needed for the coordinate identification with the normalized contact orders. All these conditions concern fixed generic-fiber objects and require the removal of only finitely many primes. The set S is chosen before the rational point s.

Lemma 4.5. *Let $s \in \mathbb{P}^1(\mathbb{Q}) \setminus \mathcal{C}$, and put $M = M_S(s)$. There is a normal extension $K_0/\mathbb{Q}$ of uniformly bounded degree splitting the fiber $\pi^{-1}(s)$, unramified outside $S \cup \{q : q \mid M\}$. Every fiber $\mathcal{A}_y$ above s is defined over K_0 and has good reduction over K_0 outside the places above the fixed set S.*

Proof. The rational fiber has $\deg \pi$ geometric points, so its splitting field is normal of degree at most $(\deg \pi)!$. Fix $q \notin S$. If the reduction of s avoids the branch sections, finite étaleness gives an unramified fiber. Otherwise it meets exactly one branch section b.

After a finite unramified local extension, all points of D_b over that reduction are rational. Formal coordinates at each give $t = z^2 e(z)$, with $e(z)$ a unit. After another unramified extension its residue has a square root; Hensel's lemma, since $q \neq 2$, gives a square root of $e(z)$ in the formal power-series ring. Replacing z by $z\sqrt{e(z)}$ gives $t = z^2$. The valuation of $t(s)$ is the normalized contact order with b. If it is even, the square roots of $t(s)$ lie in an unramified extension: remove the even uniformizer power and extract a square root of the unit after extending the residue field. Perform this at every point of D_b over the reduction. By properness every geometric point of the fiber specializes to such a point, so the whole fiber is unramified at q. This proves the ramification assertion.

Every K_0-point y extends over the local valuation rings outside S, by properness. Pulling back the spread abelian scheme proves good reduction there. This remains true at primes dividing M: those primes may ramify in the field, but introduce no bad reduction of the fiber over that field. □

Lemma 4.6. *For $A = \mathcal{A}_y$ as in Lemma 4.5, there is a normal extension $L/\mathbb{Q}$ containing K_0, of uniformly bounded degree, over which all geometric homomorphisms between all Galois conjugates of A are defined. It is unramified outside $S \cup \{q : q \mid M\}$.*

Proof. There are boundedly many conjugates, all defined over K_0. Let $\mathcal{H}$ be the direct sum of the geometric Hom groups over all ordered pairs of these conjugates. It is free abelian of uniformly bounded rank: for each coefficient prime ℓ, the tensor product of each Hom group with $\mathbb{Z}_\ell$ embeds in the homomorphisms between rank-four Tate modules, so its rank is at most 16. Galois transport acts on $\mathcal{H}$, permuting summands according to the conjugate varieties.

The image of G_{K_0} is finite, because a finite set of Hom generators is defined over a finite extension. A finite subgroup of $\mathrm{GL}_r(\mathbb{Z})$ has bounded order for bounded r; for example reduction modulo 3 is injective on it, as the principal congruence kernel is torsion-free. Intersect the kernel of the full transport action with G_{K_0}. This intersection is normal in $G_\mathbb{Q}$ and has uniformly

bounded index. Its fixed field L is normal over $\mathbb{Q}$, contains K_0, has uniformly bounded degree, and defines all the indicated Hom groups.

At a prime outside $S \cup \{q : q \mid M\}$, the field K_0 is unramified and every conjugate has good reduction. Inertia acts trivially on Tate modules for a coefficient prime different from the residue characteristic, and hence on the Hom groups embedded between them. Thus L/K_0 introduces no ramification there. □

In particular, for constants independent of s,

$$[L : \mathbb{Q}] \le d_*, \qquad \operatorname{rad}(D_L) \le C_S M, \tag{4.2}$$

where D_L is the absolute discriminant. The variety A/L has good reduction outside the places above the fixed set S.

4.5. **The effective arithmetic estimate.** We verify the exact endomorphism and compatibility hypotheses of the external height theorem. An abelian surface admitting multiplication by the division algebra B has the following dichotomy. If it is absolutely simple, its endomorphism division algebra acts on its four-dimensional rational homology, so has dimension at most four. It contains B, and therefore equals B. If it is not absolutely simple, it is isogenous to a product of elliptic curves. Those curves must be isogenous to each other, since a product of commutative endomorphism fields cannot contain B. Their common elliptic factor must have CM: otherwise the endomorphism algebra would be $M_2(\mathbb{Q})$, which cannot contain the nonsplit four-dimensional algebra B. Thus the alternatives are

$$\operatorname{End}^0_{\mathbb{Q}}(A) = B \quad \text{or} \quad A \text{ has CM.}$$

In the non-CM case, fix a quadratic subfield $K \simeq \mathbb{Q}(\sqrt{210})$ of B. It embeds by the quadratic embedding criterion, since every finite ramified prime of B ramifies in K. Its degree is $2 = \dim A$. For each $\sigma \in G_{\mathbb{Q}}$, choose a geometric isogeny $\nu_\sigma : {}^\sigma A \to A$. The map

$$b \longmapsto \nu_\sigma \, {}^\sigma b \, \nu_\sigma^{-1}$$

is a $\mathbb{Q}$-algebra automorphism of B. By Skolem–Noether it is conjugation by some $c_\sigma \in B^\times$. Consequently $c_\sigma^{-1}\nu_\sigma$ intertwines the full B-actions. Multiplying by an integer makes it an actual isogeny μ_σ satisfying

$$\mu_\sigma \, {}^\sigma f = f \mu_\sigma \qquad (f \in K). \tag{4.3}$$

All these homomorphisms and all geometric endomorphisms are defined over the field L of Lemma 4.6. There is no further field extension implicit in the adjustment.

Set $U = \operatorname{Spec} \mathcal{O}_L[S^{-1}]$ and regard A as an abelian scheme over U. Generic-fiber endomorphisms extend uniquely over U, since this abelian scheme is its Néron model there. Hence the chosen field K lies in the endomorphism algebra of the abelian scheme itself. Equation (4.3) is exactly the definition of GL_2-type with $G_{\mathbb{Q}}$-isogenies in [12, Section 7.1.1]; normality of L and definition of all endomorphisms and isogenies over L give their condition $(*)$.

Write $N_U = \prod_{\mathfrak{p} \notin U} N\mathfrak{p}$. Its prime support is contained in the fixed set S, so (4.2) implies

$$\operatorname{rad}(D_L N_U) \le C_S M.$$

Proposition 7.13 of [12], applied with $g = 2$, gives in the non-CM case

$$h_F(A) \le (8[L : \mathbb{Q}])^{288[L:\mathbb{Q}]} \operatorname{rad}(D_L N_U)^{24} \le C_1 M^{24}.$$

The last constant is uniform because $[L : \mathbb{Q}]$ is bounded. This proposition imposes no totally-real or odd-degree condition on L.

In the CM case, all geometric endomorphisms are already defined over L. Proposition 7.14 of [12] gives

$$h_F(A) \le 6^{100} \operatorname{rad}(D_L)^{10} \le C_2 M^{10} \le C_2 M^{24}.$$

Thus both cases give the same sufficient uniform exponent.

Completion of the proof of Theorem 4.1. For any rational $s \notin \mathcal{C}$, choose y above it. The preceding estimates and Lemma 4.4 give, since $M = M_S(s) \geq 1$,

$$h(s) = h(\pi(y)) \leq C(1 + \max\{0, h_F(\mathcal{A}_y)\}) \leq H_0 M^{24}.$$

The configuration, family, exceptional set, and constants were fixed before s. Taking $S_0 = S$ proves the full statement. □

The theorem supplies H for one fixed five-point configuration, which is its required scope. Neither explicit branch coordinates nor numerical values of the constants are needed for that scope. The argument uses existing canonical-model, theta-height, and effective Shafarevich results, with their hypotheses checked above; it does not establish a new unrestricted modularity theorem or an effective Mordell theorem for arbitrary curves.

5. Compatible prime patterns: Input R

This section supplies Input R without using P, E, or H. The contact configuration is arbitrary and fixed; in particular, the result applies to the configuration constructed in the proof of H. The argument is the prime-pattern construction of [4, Section 6], with the uniform estimates and the order of choices included here.

Fix five distinct points of $\mathbb{P}^1(\mathbb{Q})$, represented by primitive integral forms

$$L_0(U,V) = V, \qquad L_i(U,V) = \alpha_i U + \beta_i V \quad (1 \leq i \leq 4), \qquad \alpha_i \neq 0.$$

Thus $\alpha_i\beta_j - \alpha_j\beta_i \neq 0$ when $i \neq j$. Retain

$$F = \mathbb{Q}(\sqrt{2}), \quad d = 75 - 53\sqrt{2}, \quad E : y^2 = x^3 - d^2 x,$$

and let σ be the nontrivial automorphism of $F/\mathbb{Q}$.

Proposition 5.1. *There is a finite set $\mathcal{K} \subset \mathbb{Z}_{>0}$ such that, for every $A \in \mathbb{Z}$ and every $B > 0$, there are integers u_1, u_2, u_3, v with $v \neq 0$ and $Av = u_1 + u_2 + u_3$, for which all thirteen entries*

$$v, \qquad L_i(u_j, v) \quad (1 \leq i \leq 4,\ 1 \leq j \leq 3)$$

have the form $\epsilon\kappa r$, where $\epsilon \in \{1, -1\}$, $\kappa \in \mathcal{K}$, and $r > B$ is prime. Every such prime splits in F and gives good reduction for E and E^σ. At either place above r, their reduction-group cardinalities d_1, d_2 satisfy

$$\gcd(d_1, d_2) \mid 4, \qquad r \nmid d_1 d_2.$$

The set $\mathcal{K}$ is independent of both A and B.

The main issue is simultaneous compatibility: all thirteen prime values must occur in the same linear pattern. We obtain a lower bound for the patterns, remove the patterns containing an incompatible prime, and show that the removal costs less than the lower bound. A fixed cutoff L removes small common factors of the reduction orders. Its effect on the lower bound is only a power of $\log L$, whereas the remaining bad-prime contribution has a factor $1/L$.

5.1. Reduction orders at the selected primes.

We restrict to primes

$$r \equiv 1 \pmod 8, \qquad r \equiv 3 \pmod 7, \tag{5.1}$$

outside a fixed finite set. Such a prime splits in F, and quadratic reciprocity gives $(7/r) = -1$. Since $dd^\sigma = 7$, the reductions of d and d^σ have opposite squareclasses at either place above r. The two reduced curves are therefore the two quadratic twists of $E_0 : y^2 = x^3 - x$ over $\mathbb{F}_r$, and their Frobenius traces are opposite.

For completeness, the relevant CM trace description uses only the standard endomorphism and degree identities for elliptic curves [5, Chapters III and V]. The order-four automorphism $(x, y) \mapsto (-x, \iota y)$, where $\iota^2 = -1$ in $\mathbb{F}_r$, commutes with Frobenius. Its centralizer in the rational geometric endomorphism algebra is $\mathbb{Q}(i)$. One can check the dimension bound on the 3-adic Tate module: $T^2 + 1$ is irreducible over $\mathbb{Q}_3$, so its matrix centralizer has dimension two, and the endomorphism action is faithful after extending coefficients. Frobenius is consequently an algebraic integer $a + bi \in \mathbb{Z}[i]$, with

$$a^2 + b^2 = r, \qquad a \neq 0.$$

The orders, in either order, are

$$d_1, d_2 = r + 1 \pm 2a. \tag{5.2}$$

Both are divisible by four, because the reduced curves have full rational two-torsion. Their sum $2(r+1)$ has 2-adic valuation two, so the 2-part of their gcd is exactly four. Also $r \nmid d_i$: the elementary point-count bound $d_i \le 2r + 1 < 4r$, together with $4 \mid d_i$, excludes divisibility by the odd prime r.

If an odd prime ℓ divides both orders, then

$$a \equiv 0 \pmod{\ell}, \qquad b^2 \equiv -1 \pmod{\ell}, \qquad \ell \mid r + 1. \tag{5.3}$$

In particular $\ell \equiv 1 \pmod 4$. We impose $r \not\equiv -1 \pmod{\ell}$ for small ℓ, and count the remaining possibilities using the two simultaneous conditions on a, b.

5.2. **Fixing the multiplier catalogue before choosing A.** Write $T = (u_1, u_2, v)$ and set $u_3 = Av - u_1 - u_2$. Denote the thirteen homogeneous forms by $F_\nu(T)$, $1 \le \nu \le t = 13$. Besides v, their coefficient vectors are

$$(\alpha_i, 0, \beta_i), \quad (0, \alpha_i, \beta_i), \quad (-\alpha_i, -\alpha_i, \alpha_i A + \beta_i).$$

All are primitive. There is a threshold independent of A above which every pair remains linearly independent modulo a prime. Within each family this follows from the fixed determinants of the L_i; the terms containing A cancel in the third family. Between families it follows from the distinct first-two-coordinate directions, and independence from v follows from $\alpha_i \neq 0$.

Let S_c contain all primes up to a sufficiently large fixed threshold, including $2, 7$, all exceptions just described, and the fixed bad primes of the curves. We take that threshold larger than $2t$. Choose an integer h_0 with $t2^{-h_0} < 1$. For any A and $p \in S_c$, each primitive form vanishes modulo p^{h_0} on a fraction p^{-h_0} of the residue vectors. A union bound therefore gives a vector class on which none of the thirteen forms vanishes modulo p^{h_0}. Every lift of that class modulo p^{h_0+3} has

$$0 \le v_p(F_\nu(T)) < h_0 \qquad (1 \le \nu \le t).$$

We next construct a catalogue that anticipates every possible such choice. Put $m_2 = 8$ and $m_p = p$ for odd $p \in S_c$, and prescribe unit residues

$$c_2 = 1 \pmod 8, \qquad c_7 = 3 \pmod 7, \qquad c_p = 1 \pmod p \quad (p \in S_c \setminus \{2, 7\}).$$

A catalogue entry consists of

$$e_{p\nu} \in \{0, \dots, h_0 - 1\}, \qquad \xi_{p\nu} \in (\mathbb{Z}/m_p\mathbb{Z})^* \quad (p \in S_c, 1 \le \nu \le t).$$

For each entry put $k_{0\nu} = \prod_{p \in S_c} p^{e_{p\nu}}$. By the Chinese remainder theorem and Dirichlet's theorem [13], choose distinct primes $h_\nu \notin S_c$ satisfying

$$h_\nu \equiv c_p^{-1} \xi_{p\nu} \pmod{m_p} \quad (p \in S_c).$$

Choose one such tuple for each catalogue entry and let $\mathcal{K}$ be the union of all numbers $k_{0\nu} h_\nu$ so obtained. This is a fixed finite set. No value of A has yet been selected.

Now fix A. Choose an open rectangular box $\mathcal{B} \subset \mathbb{R}^3$ whose closure avoids the thirteen zero planes, and write ϵ_ν for the sign of F_ν on this box. For each $p \in S_c$, select a class modulo p^{h_0+3} as above and let $e_{p\nu}$ be the constant valuation on that class. The class also determines the unit

$$\xi_{p\nu} = \epsilon_\nu \frac{F_\nu(T)}{p^{e_{p\nu}}} \left(\frac{k_{0\nu}}{p^{e_{p\nu}}} \right)^{-1} \pmod{m_p}.$$

There is sufficient precision even at 2, since $h_0 + 3 - e_{p\nu} \ge 4$. Select the corresponding catalogue tuple and put $k_\nu = k_{0\nu} h_\nu$.

At $p = h_\nu$, impose $F_\nu(T) = 0$ modulo p. On this plane each other F_j is nonconstant. The conditions $\epsilon_j F_j(T) \not\equiv 0, -k_j$ remove at most $2(t-1)p < p^2$ points, so they can all be satisfied. Lift the resulting vector modulo p^2 so that

$$\epsilon_\nu F_\nu(T)/p \not\equiv 0, -k_{0\nu} \pmod p.$$

Such a lift exists because the linear form is primitive and therefore its divided value varies through all residues modulo p.

Combining these classes gives a class modulo

$$W_0 = \prod_{p \in S_c} p^{h_0+3} \prod_{\nu=1}^{t} h_\nu^2$$

on which the divided forms

$$\Psi_\nu(T) = \epsilon_\nu F_\nu(T)/k_\nu$$

are integral and nonzero modulo every prime dividing W_0. At $p \in S_c$ their residue is c_p modulo m_p; at every odd $p \mid W_0$ they also avoid -1. Notice that W_0/k_ν is divisible by every prime dividing W_0. These assertions follow respectively from the chosen unit residues, the lifted conditions at the h_ν, and the surplus powers in W_0.

All these choices are now fixed for this A. In particular, they precede the cutoff L below. Moreover W_0 ranges in a finite catalogue as A varies. On $X\mathcal{B}$ there are fixed constants $0 < c_A < C_A$ with

$$c_A X \le \Psi_\nu(T) \le C_A X \qquad (1 \le \nu \le t). \tag{5.4}$$

5.3. **A lower bound whose coefficient is uniform in the cutoff.** Fix L larger than every prime dividing W_0, and put

$$W = W_0 \prod_{\substack{p \le L \\ p \nmid W_0}} p.$$

At each new prime allow all vectors satisfying

$$\epsilon_\nu F_\nu(T) \not\equiv 0, -k_\nu \pmod{p} \qquad (1 \le \nu \le t).$$

At most $2t$ affine planes are excluded. If $\mathcal{R}_W$ denotes the allowed vector classes modulo W, including the fixed class at W_0, the Chinese remainder theorem and Mertens' estimate give

$$\frac{|\mathcal{R}_W|}{W^3} \ge W_0^{-3} \prod_{\substack{p \le L \\ p \nmid W_0}} (1 - 2t/p) \ge c_0 (\log L)^{-C_0}, \tag{5.5}$$

with positive constants independent of L. Whenever a divided form is prime, its value now avoids -1 modulo every odd prime at most L. No such prime can be a common factor in (5.2). Let $N_{A,L}(X)$ count the integer vectors in $X\mathcal{B}$ satisfying all imposed congruences for which every $\Psi_\nu(T)$ is prime.

We use the fixed finite-complexity prime-pattern theorem of Green–Tao, completed by the Möbius–nilsequence and inverse theorems [14, 15, 16], with the correction [17]. For fixed integer affine forms $\psi_1, \dots, \psi_t$ on $\mathbb{Z}^d$ with pairwise nonproportional nonzero linear parts, it gives

$$\sum_{n \in \mathcal{D}_X \cap \mathbb{Z}^d} \prod_{\nu=1}^{t} \Lambda(\psi_\nu(n)) = \operatorname{vol}(\mathcal{D}_X) \prod_p \beta_p + o(X^d) \tag{5.6}$$

on a fixed dilating box where the forms are positive, where

$$\beta_p = (1 - 1/p)^{-t} p^{-d} \#\{n \in \mathbb{F}_p^d : \psi_\nu(n) \ne 0 \ (\forall \nu)\}.$$

Pairwise nonproportionality ensures finite complexity. This application does not use a prime-tuples assertion for proportional linear parts.

Choose a smaller open box $\mathcal{B}'$ of positive volume with closure inside $\mathcal{B}$. For each allowed representative $T_0 \in [0, W)^3$, write $T = T_0 + Wn$ and use $n \in (X/W)\mathcal{B}'$. For sufficiently large X, depending on the already fixed A, L, this lies in $X\mathcal{B}$. The forms $\epsilon_\nu F_\nu(T_0 + Wn)/k_\nu$ are integral and have pairwise nonproportional linear parts. At $p \mid W$, their variable coefficients vanish and their constants are nonzero, giving $\beta_p = (1 - 1/p)^{-t} \ge 1$. At $p \nmid W$, their zero hyperplanes are distinct with pairwise intersections of codimension two, so

$$\beta_p = 1 + O_t(p^{-2}).$$

The product of these latter factors is bounded below by one positive constant independent of L, T_0, after enlarging the original fixed threshold if necessary.

Prime powers do not affect the lower bound. There are $O(X^{1/2}\log X)$ proper prime powers in the range of any one form, and each prescribed value has $O_{A,L}(X^2)$ lattice preimages. Even with the logarithmic weights this contributes $o_{A,L}(X^3)$. For genuine primes the weights are $O_A((\log X)^t)$ by (5.4). Sum (5.6) over the finitely many allowed progressions. Their volume factor W^{-3} combines with (5.5), and we obtain

$$N_{A,L}(X) \geq c_A(\log L)^{-C_0}\frac{X^3}{(\log X)^t} \tag{5.7}$$

for all sufficiently large X depending on A, L, where the positive coefficient c_A is independent of L. No uniformity in the little-oh error as L grows has been asserted or used.

5.4. **An upper sieve with explicit uniform errors.** We record the elementary sieve estimate needed twice below. For primes $p \leq Z$ in a prescribed set $\mathcal{P}_Z$, let $\Omega_p \subset \mathbb{F}_p^2$ have density $g(p) \in (0, 1/2)$. Extend g multiplicatively to squarefree integers supported on these primes. In a square with n integer choices in each coordinate, the number satisfying all forbidden congruences for $p \mid e$ is

$$n^2 g(e) + O(ne + e^2), \tag{5.8}$$

uniformly in the square's position: there are $e^2 g(e)$ relevant residue classes, each with error $O(n/e + 1)$.

Lemma 5.2. *The number of points avoiding every Ω_p is at most*

$$\frac{n^2}{G(Z)} + O(nZ^6 + Z^8), \qquad G(Z) = \sum_{\substack{e\leq Z \\ p\mid e\Rightarrow p\in\mathcal{P}_Z}}{\sum_{\text{squarefree}}}\prod_{p\mid e}\frac{g(p)}{1-g(p)}.$$

The error constant is absolute.

Proof. All indices in this proof are squarefree and supported on $\mathcal{P}_Z$. Put $g_*(p) = (1-g(p))/g(p)$ and extend multiplicatively. For weights λ_e supported on $e \leq Z$ with $\lambda_1 = 1$, the square of the sum of λ_e over the simultaneously forbidden divisors e majorizes the indicator of avoiding all forbidden sets. By (5.8), its main term, divided by n^2, is

$$Q(\lambda) = \sum_{e,f}\lambda_e\lambda_f g([e,f]) = \sum_h g_*(h)\left(\sum_{\substack{e\leq Z\\ h\mid e}}\lambda_e g(e)\right)^2.$$

The equality uses $1/g(d) = \sum_{h\mid d} g_*(h)$. Set the inner sums equal to $\mu(h)/(G(Z)g_*(h))$. Finite Möbius inversion gives

$$\lambda_e = \frac{\mu(e)}{G(Z)g(e)g_*(e)}\sum_{\substack{k\leq Z/e\\ (k,e)=1}}\frac{\mu(k)^2}{g_*(k)}.$$

Thus $\lambda_1 = 1$, $Q(\lambda) = 1/G(Z)$, and

$$|\lambda_e| \leq \prod_{p\mid e}(1-g(p))^{-1} \leq 2^{\omega(e)} \leq e.$$

Using $[e,f] \leq ef$, the total error from (5.8) is bounded by

$$O\left(n\sum_{e,f\leq Z}e^2f^2 + \sum_{e,f\leq Z}e^3f^3\right) = O(nZ^6 + Z^8).$$

□

In the applications take $Z = X^{1/100}$. Two lower bounds for $G(Z)$ will be used. If the sieving set consists of all primes above a fixed threshold and $g(p) = s/p + O_s(p^{-2})$, then

$$G(Z) \gg (\log X)^s. \tag{5.9}$$

If instead the set consists of primes $p \equiv 1 \pmod 4$ above a fixed threshold, with $g(p) = 2/p + O(p^{-2})$, then

$$G(Z) \gg \log X. \tag{5.10}$$

The latter estimate remains uniform if one additional prime is omitted. Here is a proof of the required truncation. For $y = Z^{\vartheta}$, the Euler product

$$D(y) = \prod_{\substack{p \le y \\ p \in \mathcal{P}_Z}} (1 + g_*(p)^{-1})$$

is respectively $\gg (\log y)^s$ or $\gg \log y$, by Mertens' estimates, including their fixed-progression version. Omitting one prime loses at most a factor two. In the expansion of this product, with weight $g_*(e)^{-1}$ on e, the weighted mean of $\log e$ is

$$\sum_{\substack{p \le y \\ p \in \mathcal{P}_Z}} g(p) \log p \ll \log y.$$

Choose the fixed $\vartheta > 0$ small enough. Markov's inequality then puts at least half this weight on $e \le Z$. Those terms occur in $G(Z)$, proving both bounds, including the uniformity just stated.

5.5. **Counting individual incompatible prime values.** Fix an entry ν. Let $B_{A,L}(X)$ count prime values in its range (5.4) for which (5.2) has an odd common factor $\ell > L$. We prove

$$B_{A,L}(X) \ll_A \frac{X}{L \log X} + X^{7/8} + X^{1/2} \log X, \tag{5.11}$$

with a constant independent of L.

It suffices to count integer pairs (a, b) with $|a|, |b| \le C_A \sqrt{X}$, $a \neq 0$, such that $a^2 + b^2$ is prime in the indicated range and (5.3) holds. Counting a pair several times gives an upper bound.

First take $L < \ell \le X^{1/8}$. If -1 is not a square modulo ℓ, there is nothing to count. Otherwise, for each of its two roots b_0, put

$$a = \ell U, \qquad b = b_0 + \ell V.$$

The relevant pairs fit in a square of side $n \asymp_A \sqrt{X}/\ell \gg_A X^{3/8}$. Sieve by the sufficiently large primes $p \equiv 1 \pmod 4$, $p \neq \ell$, up to Z. The change of variables is invertible modulo these primes. The equation $a^2 + b^2 = 0$ is the union of two distinct lines, so the forbidden density is exactly $2/p - 1/p^2$. Since the prime $a^2 + b^2$ has size $\asymp_A X > Z$, it avoids all these sets. Lemma 5.2 and (5.10) bound their number by

$$O_A(n^2/\log X + nZ^6 + Z^8) = O_A(X/(\ell^2 \log X)).$$

The absorption is uniform: at the shortest squares, the ratio of the error to $n^2/\log X$ is

$$O_A\left((\log X)\left(X^{6/100-3/8} + X^{8/100-3/4}\right)\right) = o_A(1).$$

Summing over these ℓ gives $O_A(X/(L \log X))$.

For $\ell > X^{1/8}$ discard primality. Because $a \neq 0$ and $\ell \mid a$, necessarily $\ell \le C_A \sqrt{X}$, and there are $O_A(\sqrt{X}/\ell)$ choices for a, with no additive constant. The two residue classes for b give $O_A(\sqrt{X}/\ell + 1)$ choices. Thus the contribution for one ℓ is $O_A(X/\ell^2 + \sqrt{X}/\ell)$. Summing even over all integers in dyadic blocks gives $O_A(X^{7/8} + \sqrt{X} \log X)$. This proves (5.11).

5.6. **Uniform counting on a prescribed prime fiber.** For a fixed prime $r \asymp_A X$, consider

$$F_\nu(T) = \epsilon_\nu k_\nu r.$$

The number of integer points $T \in X\mathcal{B}$ on this plane for which every other $\Psi_j(T)$ is prime is

$$O_A(X^2/(\log X)^{t-1}), \tag{5.12}$$

uniformly in r and independently of L. We may discard the congruences imposed in the lower-bound construction.

Since F_ν is primitive, a fixed integral unimodular coordinate change makes it the first coordinate. The other two coordinates range over a square of side $n \asymp_A X$, whose position may depend on r.

Sieve by primes $p \le Z$ above the original fixed threshold, excluding divisors of all multipliers. For sufficiently large X one has $p \neq r$, and the plane constant $c = \epsilon_\nu k_\nu r$ is nonzero modulo p.

Every other F_j restricts to a nonconstant affine function on $F_\nu = c$. Its zero line cannot coincide with the zero line of F_k for $j \neq k$: coincidence would give

$$F_j - \lambda F_k = \mu(F_\nu - c) \quad \text{for some } \lambda \in \mathbb{F}_p^*,\ \mu \in \mathbb{F}_p.$$

The left side has zero constant term, so $\mu c = 0$. Then $\mu = 0$, contradicting pairwise independence. This argument includes the case of three linearly dependent ambient forms; their restrictions may be parallel, but are still distinct.

The union of these $t-1$ zero lines has $(t-1)p + O_t(1)$ points. Consequently the forbidden density is $(t-1)/p + O_t(p^{-2})$, uniformly in the fiber. The prime values of the other divided forms exceed Z, so all these zero lines must be avoided. Lemma 5.2 and (5.9) give

$$O_A\left(X^2/(\log X)^{t-1} + XZ^6 + Z^8\right) = O_A(X^2/(\log X)^{t-1}).$$

All excluded sieving primes were fixed before r, L, and the residue error is uniform in translations of the square. This proves the claimed uniformity in (5.12).

5.7. **Completion of the proof.** For each of the t entries, multiply (5.11) by (5.12), then take a union bound. The number of prime patterns containing an incompatible prime is at most

$$\frac{C_A X^3}{L(\log X)^t} + o_A(X^3/(\log X)^t),$$

where the displayed coefficient is independent of L. Indeed the two other terms divided by $X^3/(\log X)^t$ are $O_A(X^{-1/8}\log X)$ and $O_A(X^{-1/2}(\log X)^2)$.

Choose the fixed cutoff L, after A and the initial data, so large that

$$C_A/L < \tfrac{1}{2} c_A (\log L)^{-C_0}.$$

Then choose X sufficiently large for (5.7), all the upper estimates, and the remaining error to leave a positive count. Small common odd factors were excluded by congruences; all larger ones have just been removed. The surviving primes therefore satisfy the gcd condition, and the earlier elementary bound gives $r \nmid d_1 d_2$. Finally increase X until $c_A X > B$. The identities defining u_3 give the required simultaneous representation. The only multipliers used belong to the finite catalogue fixed before A, B. This proves Proposition 5.1. □

The precise compatibility used later. At a selected prime, the congruences

$$h \equiv 0 \pmod{d_1}, \qquad h \equiv 4 \pmod{d_2 r^c}$$

are compatible for every fixed $c \ge 1$. Taking $j = d_2$ supplies the paired formal-group witnesses used by the manuscript. This is one simultaneous Chinese remainder problem, not the combination of unrelated existence statements. If e_1, e_2 denote the orders of the reduced chosen points, the weaker exact condition would be $\gcd(e_1, e_2 r^c) \mid 4$; the stronger reduction-group condition just proved guarantees it for every choice of the fixed point.

6. Square conditions and local divisibility

In this section and through Section 8, assume P. Inputs E, H, and R are the theorems just proved; there are no other unresolved arithmetic assumptions in the construction.

The pole-parity predicate can control denominators without defining an integrality condition. The useful observation is that a denominator of the form $1 + rW$ is forced to be a unit at the primes where r has odd valuation. We first give a rational version and then a version selecting one place of the quadratic field. All the conditions below are existential conditions on rational variables; valuations occur only in the statements and proofs of their properties.

Lemma 6.1. *Suppose $\Phi(r)$ and $\Phi(ru^2)$ hold, and let $q \notin S_\Phi$ have odd valuation $e = v_q(r)$. Then $e > 0$ and*

$$v_q(u) \ge -\left\lfloor \frac{e}{2} \right\rfloor.$$

Proof. The condition $\Phi(r)$ excludes a negative odd value of e. If $u \neq 0$, the valuation $e + 2v_q(u)$ is odd, so $\Phi(ru^2)$ requires it to be nonnegative. This gives the bound. The case $u = 0$ uses the convention $v_q(0) = +\infty$. □

6.1. **Several rational targets with one denominator.**

Definition 6.2. For a fixed positive integer t, define $\mathrm{Div}(r; D_1, \ldots, D_t)$, with $r, D_i \in \mathbb{Q}$, by requiring $r \neq 0$, $\Phi(r)$, and rational witnesses $W, X_1, \ldots, X_t$ such that, on writing $Y = 1 + rW$,

$$\begin{aligned} Y \neq 0, \qquad & D_i Y = rX_i \quad (1 \le i \le t), \\ \Phi(rW^2), \qquad & \Phi(rX_i^2) \quad (1 \le i \le t). \end{aligned} \tag{6.1}$$

The length t is fixed in each use of this definition, rather than a variable in the formula.

Lemma 6.3 (Soundness)**.** *If* $\mathrm{Div}(r; D_1, \ldots, D_t)$ *holds and* $q \notin S_\Phi$ *has odd* $v_q(r)$*, then*

$$v_q(r) > 0, \qquad v_q(D_i) \ge 1 \quad (1 \le i \le t).$$

Proof. Put $e = v_q(r)$ and $d = \lfloor e/2 \rfloor$. Lemma 6.1 gives $e > 0$ and valuations at least $-d$ for $W, X_1, \ldots, X_t$. Thus $v_q(rW) \ge e - d > 0$, so $Y = 1 + rW$ is a q-adic unit. Equation (6.1) gives $v_q(D_i) \ge e - d \ge 1$, including zero targets under the infinite-valuation convention. □

Lemma 6.4 (Completeness for integer moduli)**.** *Let* $m \in \mathbb{Z} \setminus \{0\}$ *and* $D_1, \ldots, D_t \in \mathbb{Q}$ *satisfy*

$$v_p(D_i) \ge v_p(m) \qquad (p \mid m,\ 1 \le i \le t). \tag{6.2}$$

Then $\mathrm{Div}(m; D_1, \ldots, D_t)$ *has witnesses* $W, X_1, \ldots, X_t$ *in* $\mathbb{Z}$*. For* $m = \pm 1$*, no condition on the targets is needed.*

Proof. Condition (6.2) permits a common denominator

$$D_i/m = A_i/B, \qquad A_i, B \in \mathbb{Z}, \quad B \neq 0, \quad \gcd(B, m) = 1.$$

Choose a nonzero integer z with $zB \equiv 1 \pmod{m}$, and set

$$W = \frac{zB - 1}{m}, \qquad X_i = zA_i.$$

Then $Y = 1 + mW = zB \neq 0$ and $D_i Y = mX_i$. All witnesses and all arguments of Φ are integers, so Input P supplies the required conditions. This works for negative m as well. When $m = \pm 1$, take $z = 1$; the same construction applies to arbitrary rational targets. □

6.2. **A selected place of the quadratic field.** For rational r, b, C, impose the root conditions

$$r \neq 0, \qquad \Phi(r), \qquad b^2 - 2 = rC, \qquad \Phi(rC^2). \tag{6.3}$$

They will be shared by all local conditions attached to one occurrence of r. The parameter b selects a residue of $\sqrt{2}$.

Definition 6.5. For a fixed integer $k \ge 1$ and $D \in F$, let $\mathrm{Loc}_{r,b,k}(D)$ assert the existence of $X_0, X_1, V, W \in \mathbb{Q}$ such that

$$\begin{aligned} X = X_0 + X_1\sqrt{2}, \qquad & Y = 1 + rW + V(\sqrt{2} - b) \neq 0, \qquad DY = r^k X, \\ & \Phi(rX_0^2), \quad \Phi(rX_1^2), \quad \Phi(rV^2), \quad \Phi(rW^2). \end{aligned} \tag{6.4}$$

The root conditions (6.3) are imposed separately. An F-equation is interpreted as two rational equations in the basis $1, \sqrt{2}$, as in Appendix B.

Lemma 6.6 (Soundness at the selected place)**.** *Assume* (6.3)*, and let* $q \notin S_\Phi \cup \{2\}$ *have odd valuation* $e = v_q(r)$*. Then* q *splits in* F*. At the place* $w \mid q$ *where* $\sqrt{2}$ *reduces to* b*, normalized by* $w(q) = 1$*,*

$$\mathrm{Loc}_{r,b,k}(D) \quad \Longrightarrow \quad w(D) \ge k.$$

The same place works for every condition using these root data.

Proof. Write $d = \lfloor e/2 \rfloor$. Lemma 6.1 gives $e > 0$ and $v_q(C) \geq -d$, whence

$$v_q(b^2 - 2) \geq e - d > 0.$$

Consequently b is a q-adic unit and 2 is a square modulo q. Since $q \neq 2$, it splits in F. At the selected place, the factor $\sqrt{2} + b$ is a unit, so

$$w(\sqrt{2} - b) = v_q(b^2 - 2) \geq e - d.$$

For witnesses in (6.4), the same lemma gives valuations at least $-d$ for X_0, X_1, V, W. Therefore

$$w(rW) \geq e - d > 0, \qquad w(V(\sqrt{2} - b)) \geq e - 2d = 1.$$

The denominator Y is a unit. Since $w(X) \geq -d$, the equation $DY = r^k X$ yields

$$w(D) \geq ke - d \geq k.$$

The last inequality uses $e \geq 1$ and $k \geq 1$; zero targets cause no exception. The selected place depends only on q and the shared value of b. □

Lemma 6.7 (Completeness at a split prime)**.** *Let r be an odd rational prime split in F, let $w \mid r$, and choose $b \in \mathbb{Z}$ reducing to $\sqrt{2}$ at w. Put $C = (b^2 - 2)/r \in \mathbb{Z}$. For every $D \in r^k \mathcal{O}_{F,w}$, the root conditions* (6.3) *hold and* $\mathrm{Loc}_{r,b,k}(D)$ *has witnesses $X_0, X_1, V, W \in \mathbb{Z}$.*

Proof. Since $\mathcal{O}_F = \mathbb{Z}[\sqrt{2}]$, write

$$D/r^k = A/B, \qquad A, B \in \mathcal{O}_F, \quad B \notin \mathfrak{p}_w.$$

Write $B = B_0 + B_1\sqrt{2}$ and $B_b = B_0 + bB_1$. The integer B_b is nonzero modulo r. Choose $z \in \mathbb{Z}$ with $zB_b \equiv 1 \pmod{r}$; then $z \neq 0$. Take

$$X = zA, \qquad V = zB_1, \qquad W = \frac{zB_b - 1}{r}.$$

The denominator in (6.4) is exactly

$$1 + rW + V(\sqrt{2} - b) = zB_b + zB_1(\sqrt{2} - b) = zB.$$

It is nonzero, and $DY = r^k X$. The coefficients of X, as well as V, W, C, are integers. Input P therefore supplies every occurrence of Φ. □

These definitions impose square conditions only on the witnesses explicitly listed. Witnesses internal to Φ, and reciprocal witnesses used to express inequations, are not subjected to further conditions. Each definition is therefore a finite existential formula. The soundness lemmas are one-way implications; the completeness lemmas state exactly which intended instances admit witnesses.

7. Multiplication from quadratic height growth

We now construct one formula for multiplication of integer indices. Its auxiliary rational values satisfy $\xi_1\xi_2 = \xi_3$. Local comparisons will show that a nonzero discrepancy $n_1n_2 - n_3$ bounds certain coordinate heights. Applying a fixed multiplication map to the points makes the quadratic lower height bound incompatible with that discrepancy. The local precision in this entire construction is the fixed integer c. Write ℓ_c for the degree-c truncation of the formal logarithm of E, as in Appendix A.

7.1. Local comparisons.

For U on E or E^σ, write

$$\mathrm{For}_{r,b}(U): \quad U = O \quad \text{or} \quad (U \neq O,\ x(U)y(U) \neq 0,\ \mathrm{Loc}_{r,b,1}(1/x(U))).$$

At a prime outside S with odd $v_q(r)$, the root conditions and Lemma 6.6 make this a sufficient condition for U to reduce to O at the selected place. We use $z(O) = 0$ on the identity branch. All other quotients are accompanied by their nonzero-denominator conditions. For nonidentity points in G or G^σ, the conditions $x, y \neq 0$ are automatic: a point with either coordinate zero is two-torsion.

Definition 7.1. Let $\boldsymbol{\xi} = (\xi_1, \xi_2, \xi_3) \in \mathbb{Q}^3$ and $\mathbf{T} = (T_1, T_2, T_3) \in G^3$. Put $\xi_0 = 1$ and $T_0 = P$. The condition $\mathrm{Cmp}(r; \boldsymbol{\xi}, \mathbf{T})$ requires root data b, C satisfying (6.3), and points

$$J, R_0, R_1, R_2, R_3 \in G, \qquad J \neq O, \qquad \ell_c(z(R_0)) \neq 0.$$

Set $V_i = R_i - [4]T_i$. For every $i = 0, 1, 2, 3$, impose

$$\begin{gathered} \mathrm{For}_{r,b}(R_i), \quad \mathrm{For}_{r,b}(J^\sigma), \quad \mathrm{For}_{r,b}(V_i^\sigma), \\ \mathrm{Loc}_{r,b,c}\left(\frac{\ell_c(z(R_i))}{\ell_c(z(R_0))} - \xi_i\right), \qquad \mathrm{Loc}_{r,b,c}\left(\frac{z(V_i^\sigma)}{z(J^\sigma)}\right). \end{gathered} \tag{7.1}$$

The local conditions share r, b, C but otherwise have separate witnesses.

Lemma 7.2. *Write $T_i = n_i P$ for $i = 1, 2, 3$.*

(1) If $\mathrm{Cmp}(r; \boldsymbol{\xi}, \mathbf{T})$ holds, then, for every prime $q \notin S$ with odd $v_q(r)$,

$$v_q(\xi_i - n_i) \geq c \quad (i = 1, 2, 3). \tag{7.2}$$

(2) If $\xi_i = n_i \in \mathbb{Z}$ and $r \notin S$ is a prime supplied by Input R, then $\mathrm{Cmp}(r; \boldsymbol{\xi}, \mathbf{T})$ holds.

Proof. For the first assertion, Lemma 6.6 selects one place $w \mid q$ at which all local conditions are sound. Write $R_i = t_i P$ and $J = jP$, where $j \neq 0$, and set $n_0 = 1$. The points tested by For lie in their formal subgroups at w. The exact valuation comparison of Lemma A.3, applied on E^σ, gives

$$v_q\left(\frac{t_i - 4n_i}{j}\right) \geq c \qquad (0 \leq i \leq 3).$$

Because j is an integer, $t_i \equiv 4n_i \pmod{q^c}$. The case $i = 0$ gives $t_0 \equiv 4 \pmod{q^c}$. Since $q \neq 2$, the denominator index is a unit and

$$t_i/t_0 \in \mathbb{Z}_q, \qquad t_i/t_0 \equiv n_i \pmod{q^c}.$$

Now $q > c + 1$, by the choice of S. The truncated-logarithm estimate in Lemma A.3 and the first local quotient in (7.1) show that $v_q(\xi_i - n_i) \geq c$. In particular every ξ_i is q-integral. The argument includes zero numerator points under the convention $z(O) = 0$.

For completeness, choose either place $w \mid r$. Let d_1, d_2 be the orders of the reductions of E, E^σ there. Input R gives $\gcd(d_1, d_2) \mid 4$ and $r \nmid d_1 d_2$, so there is a positive integer h with

$$h \equiv 0 \pmod{d_1}, \qquad h \equiv 4 \pmod{d_2 r^c}.$$

Take

$$J = d_2 P, \qquad R_i = h n_i P \quad (0 \leq i \leq 3).$$

Here $n_0 = 1$, so $R_0 = hP$, and $V_i = (h - 4)n_i P$. These choices put every R_i in the formal subgroup of E and J^σ, V_i^σ in that of E^σ. The truncated-logarithm estimate gives

$$w\left(\frac{\ell_c(z(R_i))}{\ell_c(z(R_0))} - n_i\right) \geq c,$$

and the exact valuation comparison gives

$$w\left(\frac{z(V_i^\sigma)}{z(J^\sigma)}\right) = v_r\left(\frac{(h-4)n_i}{d_2}\right) \geq c.$$

Both denominator points are nonidentity. The same formal estimate ensures $\ell_c(z(R_0)) \neq 0$. Choose one integer lift b of $\sqrt{2}$ at w, and put $C = (b^2 - 2)/r$. Lemma 6.7 supplies all conditions in (7.1), including the depth-one conditions on reciprocal x-coordinates, with this shared choice. Identity points use their identity branches. □

7.2. **Representations and the discrepancy bound.**

Definition 7.3. The condition $\mathrm{Rep}(A;\boldsymbol{\xi},\mathbf{T})$ requires $u_1,u_2,u_3,v\in\mathbb{Q}$ with $v\neq 0$ and

$$Av=u_1+u_2+u_3.$$

For each of the thirteen entries

$$v,\qquad L_i(u_j,v)\quad (1\le i\le 4,\ 1\le j\le 3), \tag{7.3}$$

require an expression $\epsilon\kappa r$, with $\epsilon\in\{1,-1\}$, $\kappa\in\mathcal{K}$, $r\in\mathbb{Q}^\times$, and impose $\mathrm{Cmp}(r;\boldsymbol{\xi},\mathbf{T})$. Different occurrences have separate witnesses. The finite choices of ϵ,κ are finite disjunctions.

Lemma 7.4. *For every $A,n_1,n_2,n_3\in\mathbb{Z}$, the condition $\mathrm{Rep}(A;\boldsymbol{\xi},\mathbf{T})$ holds with $\xi_i=n_i$ and $T_i=n_iP$.*

Proof. Apply Input R to A, taking every selected prime outside the fixed finite set S. Apply the completeness assertion of Lemma 7.2 to each occurrence in (7.3). □

Lemma 7.5 (A height bound from a nonzero discrepancy)**.** *Suppose $\xi_1\xi_2=\xi_3$, $T_i=n_iP$, and $\delta=n_1n_2-n_3\neq 0$. Then*

$$\mathrm{Rep}(A;\boldsymbol{\xi},\mathbf{T})\quad\Longrightarrow\quad h(A)\le 3H|\delta|+2. \tag{7.4}$$

Proof. Let $s_j=(u_j:v)$ be one of the three points in a representation. All five defining linear forms are nonzero there, so $s_j\notin\mathcal{C}$. Suppose q occurs in $M_S(s_j)$, and put $\mu=\min\{v_q(u_j),v_q(v)\}$. The minimum valuation among the five linear-form values is μ: their coefficients are integral, and a pair with q-unit determinant recovers u_j,v. Since one normalized contact order is odd, the valuations of these five values cannot all have the same parity. At least one is odd. Its expression $\epsilon\kappa r$ has odd $v_q(r)$, since the prime divisors of every $\kappa\in\mathcal{K}$ lie in S.

Lemma 7.2 gives $\xi_i\in\mathbb{Z}_q$ and $v_q(\xi_i-n_i)\ge c$ for all three indices. The identity

$$\delta=(n_1-\xi_1)n_2+\xi_1(n_2-\xi_2)+(\xi_3-n_3)$$

then gives $v_q(\delta)\ge c$. Thus

$$M_S(s_j)^c\mid\delta.$$

The height estimate (2.7) implies $h(s_j)\le H|\delta|$. Finally $A=u_1/v+u_2/v+u_3/v$, so

$$h(A)\le\sum_{j=1}^{3}h(s_j)+\log 3\le 3H|\delta|+2.$$

□

7.3. **The multiplication formula.** Fix a positive integer L such that

$$L^2>18a_*H+a_*^2+8a_*. \tag{7.5}$$

All constants are now fixed. Multiplication by L on the elliptic curve is a fixed algebraic operation, obtained by finitely many group additions. It does not use multiplication of variable integer indices.

Definition 7.6. For point codes T_1,T_2,T_3, require $T_i\in G$ and rational ξ_1,ξ_2,ξ_3 with $\xi_1\xi_2=\xi_3$. For each $j=1,2,3$, require either $T_j=O$, or witnesses $Q_j\in G\setminus\{O\}$ and $A_{j,0},A_{j,1},A_{j,2}\in\mathbb{Q}$ satisfying

$$\begin{gathered}Q_j=[L]T_j,\qquad A_{j,2}\neq 0,\qquad x(Q_j)=\frac{A_{j,0}+A_{j,1}\sqrt{2}}{A_{j,2}},\\ \mathrm{Rep}(A_{j,t};\boldsymbol{\xi},\mathbf{T})\quad (t=0,1,2).\end{gathered} \tag{7.6}$$

Call the resulting condition $\mathrm{Mul}_G(T_1,T_2,T_3)$. Every representation in (7.6) compares the full original tuple $\mathbf{T}$, not the amplified points Q_j.

Membership in $G=m_0E(F)$, fixed group operations, and the coordinate conditions are existential by Appendix B. All other predicates in the definition have fixed finite length. Consequently Mul_G is one existential formula over $\mathbb{Q}$.

Theorem 7.7 (Multiplication on the cyclic group)**.** *For all $n_1, n_2, n_3 \in \mathbb{Z}$,*

$$\mathbb{Q} \models \mathrm{Mul}_G(n_1P, n_2P, n_3P) \iff n_1n_2 = n_3. \tag{7.7}$$

Together with the elliptic group law, this gives a Diophantine model of the ring $\mathbb{Z}$ on the rational point codes of the original group G, under $n \mapsto nP$.

Proof. If $n_1n_2 = n_3$, take $\xi_i = n_i$. For every nonidentity $T_j = n_jP$, the point $Q_j = [L]T_j$ is nonidentity, since G is torsion-free. Its x-coordinate has a representation in (7.6) with all three $A_{j,t}$ integral. Lemma 7.4 supplies the required witnesses. For $T_j = O$, use the identity branch. This proves completeness for positive, negative, and zero indices.

Conversely, suppose the formula holds with $T_i = n_iP$ and set $\delta = n_1n_2 - n_3$. Assume $\delta \neq 0$. Then

$$B = \max\{|n_1|, |n_2|, |n_3|\} \geq 1, \qquad |\delta| \leq B^2 + B \leq 2B^2.$$

Choose j with $|n_j| = B$. Its branch supplies $Q_j = Ln_jP$ and three coordinate representatives. Lemma 7.5 and the height inequalities over F give

$$\begin{aligned} h(x(Q_j)) &\leq \sum_{t=0}^{2} h(A_{j,t}) + h(\sqrt{2}) + \log 2 \\ &\leq 9H|\delta| + 8 \leq 18HB^2 + 8. \end{aligned} \tag{7.8}$$

Here $h(\sqrt{2}) + \log 2 = \frac{3}{2}\log 2 < 2$. On the other hand, (2.8) gives

$$h(x(Q_j)) \geq \frac{L^2B^2}{a_*} - a_*.$$

Combining these inequalities yields

$$\left(\frac{L^2}{a_*} - 18H\right) B^2 \leq a_* + 8,$$

contrary to (7.5) and $B \geq 1$. Thus $\delta = 0$, proving (7.7).

The set of point codes for G and its addition law are existentially defined; O represents zero and P represents one. The computable bijection $n \mapsto nP$ therefore transports both ring operations to Diophantine relations on G. □

Remark 7.8. The auxiliary rational values need not be the integer indices. For $T_1 = T_2 = T_3 = O$, every coordinate condition takes its identity branch, and

$$(\xi_1, \xi_2, \xi_3) = (1/2, 1/2, 1/4)$$

satisfies the remaining equation $\xi_1\xi_2 = \xi_3$. These nonintegral witnesses represent the true index relation $0 \cdot 0 = 0$; they do not evaluate the indices as rational scalars.

7.4. **Polynomial and computably enumerable relations.**

Theorem 7.9. *For every $f \in \mathbb{Z}[X_1, \ldots, X_r]$, one can effectively construct an existential formula $\Theta_f(T_1, \ldots, T_r)$ over $\mathbb{Q}$ such that*

$$\mathbb{Q} \models \Theta_f(n_1P, \ldots, n_rP) \iff f(n_1, \ldots, n_r) = 0.$$

The formula requires all its free point arguments to belong to G.

Proof. Choose an arithmetic circuit for f. Introduce a point in G for each gate. An integer constant d is represented by the fixed point $[d]P$; an addition gate uses the group law; a multiplication gate uses Mul_G. Require the output to be O. Induction through the circuit shows that the index at each gate is the value of its corresponding integer expression. Thus the output condition is equivalent to $f(\mathbf{n}) = 0$ for the prescribed input indices. Conversely, those integer gate values supply point witnesses whenever $f(\mathbf{n}) = 0$.

The point $[d]P$ is computable from d, and each gate uses a fixed existential relation. This makes the construction effective once the fixed arithmetic data have been chosen. Constant polynomials are covered by constant output gates, while the input point conditions remain in force. □

Corollary 7.10. *For every computably enumerable set $A \subseteq \mathbb{Z}^r$, the relation*

$$\widehat{A} = \{(n_1P, \dots, n_rP) : (n_1, \dots, n_r) \in A\}$$

is Diophantine over $\mathbb{Q}$.

Proof. The Davis–Putnam–Robinson–Matiyasevich theorem [18, Theorem 8.1] gives a polynomial representation over the integers. To spell out the signed-integer convention, apply its nonnegative-integer version to the computably enumerable set

$$\{(\mathbf{u}, \mathbf{v}) \in \mathbb{N}^r \times \mathbb{N}^r : \mathbf{u} - \mathbf{v} \in A\}.$$

The positive and nonnegative versions differ only by a shift. Impose $n_i = u_i - v_i$ and express each nonnegative variable by a sum of four squares of unrestricted integers. Combining the resulting finite system by a sum of squares gives an integer polynomial $U(\mathbf{n}, \mathbf{w})$ with

$$\mathbf{n} \in A \quad \Longleftrightarrow \quad \exists \mathbf{w} \in \mathbb{Z}^s \; U(\mathbf{n}, \mathbf{w}) = 0.$$

Use Theorem 7.9 for U, retain the point arguments representing $\mathbf{n}$, and existentially quantify the point arguments representing $\mathbf{w}$. □

The arithmetic inputs and the multiplication predicate are fixed throughout. The formula for a particular f or A may have a different number of gate or witness variables; no variable-length list occurs inside any one formula.

8. Recovering the integer represented by a point

Theorem 7.7 supplies arithmetic on the points nP. We now recover the corresponding scalar $n \in \mathbb{Q}$. The argument combines a height bound with a congruence: if $x \neq n$ and $v_3(x - n) \geq k$, then

$$k \log 3 \leq h(x - n) \leq h(x) + h(n) + \log 2.$$

Thus a bound for $h(x)$ in terms of the represented integer allows a sufficiently precise congruence to force equality. We construct both conditions by fixed existential formulas. The exceptional set S and the positive integers H, c are those fixed in Section 2.

Throughout this section use the rational coefficient projection

$$\rho : F \longrightarrow \mathbb{Q}, \qquad \rho(a_0 + a_1\sqrt{2}) = a_0 = \frac{1}{2}\operatorname{Tr}_{F/\mathbb{Q}}(a_0 + a_1\sqrt{2}).$$

The same notation denotes its extension to $F \otimes_{\mathbb{Q}} \mathbb{Q}_p = \prod_{w|p} F_w$. This is a continuous $\mathbb{Q}_p$-linear map. As before, $z(O) = 0$.

8.1. **A support condition.** A nonidentity point of G has nonzero x- and y-coordinates: on the displayed equation a point with either coordinate zero has order two, whereas G is torsion-free. The rational functions below are therefore defined on their stated domains.

Definition 8.1. For $r \in \mathbb{Q}$ and a point code T, let $\operatorname{Supp}(r, T)$ assert that $r \neq 0$, $T \in G \setminus \{O\}$, and there exist $J, R \in G \setminus \{O\}$ and $a, b, D \in \mathbb{Q}$ such that

$$\operatorname{Mul}_G(T, J, R), \qquad \frac{1}{x(J)} = a + b\sqrt{2}, \qquad D = \rho\left(\frac{z(R)}{z(J)}\right), \qquad \operatorname{Div}(r; a, b, D). \tag{8.1}$$

All rational functions are encoded together with their nonzero denominators.

This is one finite existential formula. If $T = mP$ and $J = jP$, its multiplication condition says $R = mjP = [m]J$.

Proposition 8.2. *The support relation has the following two properties.*

(1) If $m \in \mathbb{Z} \setminus \{0\}$ and $\operatorname{Supp}(r, mP)$ holds, then, for every prime $q \notin S$,

$$v_q(r) \text{ odd} \quad \Longrightarrow \quad v_q(r) > 0 \text{ and } q \mid m. \tag{8.2}$$

(2) For every $m \in \mathbb{Z} \setminus \{0\}$,

$$\operatorname{Supp}(m, mP). \tag{8.3}$$

Proof. For the first assertion, fix $q \notin S$ with $v_q(r)$ odd. Lemma 6.3 gives

$$v_q(r) > 0, \qquad v_q(a), v_q(b), v_q(D) \geq 1.$$

At every place $w \mid q$, normalized by $w(q) = 1$, the two coefficient bounds imply $w(1/x(J)) \geq 1$. The curve has good reduction there, so J lies in the formal subgroup. Its multiple $R = [m]J$ lies in that subgroup as well. By Lemma A.1,

$$[m](Z) = mZ + Z^2 F_m(Z), \qquad F_m(Z) \in \mathcal{O}_{F_w}[[Z]]$$

for every integer m, including negative integers. Since $w(z(J)) \geq 1$, it follows that

$$w\left(\frac{z(R)}{z(J)} - m\right) \geq 1.$$

These inequalities hold at every place above q. Because $q \neq 2$, applying ρ gives $v_q(D - m) \geq 1$. Together with $v_q(D) \geq 1$, this implies $q \mid m$.

For the second assertion, first suppose $|m| > 1$. At each place above each prime $p \mid m$, choose a sufficiently small open formal subgroup about O. For the fixed integer m, the quotient

$$u_m(J) = \frac{z([m]J)}{z(J)}$$

extends analytically to O with value m, by the same multiplication series. Also $1/x(J)$ tends to zero as J tends to O. The two rational coefficient projections and ρ are continuous on $F \otimes_{\mathbb{Q}} \mathbb{Q}_p$. Consequently the subgroups can be chosen so that, whenever J lies in all the chosen neighborhoods above p,

$$v_p(a), v_p(b), v_p(\rho(u_m(J)) - m) \geq v_p(m).$$

These are simultaneous conditions at all places above p. They apply also at bad and ramified primes: the integral formal group and its multiplication series are available sufficiently close to O, as recorded in Lemma A.1. Fixed denominators in the coefficient projections merely require smaller neighborhoods.

Each chosen open subgroup has finite index in the compact local group $E(F_w)$. A common nonzero integer multiple $J = jP$ therefore belongs to all of them. Both J and $R = [m]J$ are nonidentity, and for the resulting rational quantities we have

$$v_p(a), v_p(b), v_p(D) \geq v_p(m) \qquad (p \mid m).$$

Lemma 6.4, applied to the triple (a, b, D) with modulus m, supplies the required witnesses. The model multiplication condition holds because $R = [m]J$.

For $m = \pm 1$, choose any nonidentity $J \in G$ and put $R = [m]J$. The local hypotheses of Lemma 6.4 are empty for these moduli, so it again supplies the triple certificate. □

The support condition concerns odd valuations outside S. It is not an equality test: $\mathrm{Supp}(1, mP)$ holds for every nonzero integer m, by the same argument with modulus 1 and any nonidentity J. Only the two properties in Proposition 8.2 will be used.

8.2. **Congruences with variable precision.** We use the fixed existential definition of $\mathbb{Z}_{(3)} = \{u \in \mathbb{Q} : v_3(u) \geq 0\}$ supplied by Lemma C.1. To produce a congruence of variable precision, we first choose one fixed formal point.

The prime 3 is inert and unramified in F. Moreover,

$$\mathrm{N}_{F/\mathbb{Q}}(d) = 75^2 - 2 \cdot 53^2 = 7, \qquad \Delta(E) = 64d^6,$$

so the displayed equation has good reduction at the unique place $w \mid 3$. Write $\mathbb{F}_9 = \mathbb{F}_3(\alpha)$ with $\alpha^2 = 2$. Since $d \equiv \alpha$, the reduced equation is

$$y^2 = x^3 + x.$$

The map $x \mapsto x^3 + x$ is the nonzero $\mathbb{F}_3$-linear trace $\mathbb{F}_9 \to \mathbb{F}_3$, with three preimages of each value. The values $0, 1, 2$ give respectively $3, 6, 6$ affine points: each nonzero element of $\mathbb{F}_3$ is a square in $\mathbb{F}_9$, since $1 = 1^2$ and $2 = \alpha^2$. Including the point at infinity gives

$$\#\widetilde{E}(\mathbb{F}_9) = 3 + 6 + 6 + 1 = 16.$$

Thus the fixed nonidentity point $U = [16]P$ lies in the formal subgroup at w. Normalize $w(3) = 1$.

For $n \in \mathbb{Z}$ and $k \in \mathbb{N}$, put

$$B_k = [3^k]U = [16 \cdot 3^k]P, \qquad A_{n,k} = [n]B_k, \tag{8.4}$$

and define rational numbers

$$c_{n,k} = \rho\left(\frac{z(A_{n,k})}{z(B_k)}\right), \qquad b_k = \rho\left(\frac{z(B_k)}{z(U)}\right). \tag{8.5}$$

The points B_k are nonidentity. The convention $z(O) = 0$ applies when $n = 0$.

Lemma 8.3. *For every $n \in \mathbb{Z}$ and $k \in \mathbb{N}$,*

$$v_3(c_{n,k} - n) \geq k + 1, \qquad v_3(b_k) = k. \tag{8.6}$$

In particular $b_k \neq 0$ and $b_0 = 1$.

Proof. Let $\mathcal{O}_w$ be the valuation ring of F_w, and let ℓ be the formal logarithm. Lemma A.2 gives $\ell(Z)/Z \in 1 + 3\mathcal{O}_w$ for every nonzero formal parameter Z at w. The logarithm identity $\ell(z(B_k)) = 3^k \ell(z(U))$ therefore gives

$$\frac{z(B_k)}{z(U)} = 3^k \frac{\ell(z(U))/z(U)}{\ell(z(B_k))/z(B_k)} \in 3^k(1 + 3\mathcal{O}_w).$$

The projection ρ maps $1 + 3\mathcal{O}_w$ into $1 + 3\mathbb{Z}_3$, since it is the trace divided by the 3-adic unit 2. This proves $v_3(b_k) = k$. The case $k = 0$ also gives $b_0 = \rho(1) = 1$. The same identity shows

$$w(z(B_k)) = w(z(U)) + k \geq k + 1.$$

For any integer n, the integral multiplication series of Lemma A.1 gives

$$w\left(\frac{z(A_{n,k})}{z(B_k)} - n\right) \geq w(z(B_k)) \geq k + 1.$$

Taking the trace divided by 2 preserves this bound. This proves the first assertion for positive, negative, and zero n. □

Proposition 8.4. *There is a fixed existential formula $\mathrm{Cong}_3(x, T, K)$ such that*

$$\mathrm{Cong}_3(x, nP, kP) \iff v_3(x - n) \geq k \qquad (x \in \mathbb{Q},\ n \in \mathbb{Z},\ k \in \mathbb{N}). \tag{8.7}$$

The formula is false if either point argument is not a valid code for an element of G, or if the index of K is negative.

Proof. The relation

$$\{(k, \nu) \in \mathbb{Z}^2 : k \geq 0,\ \nu = 16 \cdot 3^k\}$$

is computable. Use its fixed definition on G^2 from Corollary 7.10 to obtain $B = B_k$ from $K = kP$, and use $\mathrm{Mul}_G(T, B, A)$ to obtain $A = A_{n,k}$. Introduce c, b by the rational-function identities in (8.5). Finally require the existence of $u \in \mathbb{Z}_{(3)}$ with

$$x - c = bu.$$

Membership in this fixed local ring is existential by Lemma C.1, and $b \neq 0$ by Lemma 8.3. Thus the last condition is equivalent to $v_3(x - c_{n,k}) \geq k$. Since $v_3(c_{n,k} - n) \geq k + 1$, it is equivalent to $v_3(x - n) \geq k$.

Require both free point codes to lie in G. The graph used to choose B has $k \geq 0$ as part of its definition, so the stated domain restrictions follow. Exponentiation appears only in this one fixed relation on integer indices; no variable exponent occurs in a ring term of the formula. □

8.3. **Height control and scalar evaluation.** Write the four finite defining forms from Section 2 as $L_i(U,V) = \alpha_i U + \beta_i V$, with $\alpha_i, \beta_i \in \mathbb{Z}$ and $\alpha_i \neq 0$. For $n \in \mathbb{Z}$ put

$$m_i(n) = \alpha_i n + \beta_i, \qquad R(n) = \prod_{i=1}^{4} m_i(n), \qquad \kappa(n) = HR(n)^{2c} + n^2 + 1. \tag{8.8}$$

The product $R(n)$ is signed. The function κ is an integer-coefficient polynomial and is positive on every integer. By Theorem 7.9, its graph on G is defined by the fixed formula $\Theta_{Y-\kappa(X)}(T,K)$:

$$\Theta_{Y-\kappa(X)}(nP, kP) \quad \Longleftrightarrow \quad k = \kappa(n).$$

This formula uses only the fixed arithmetic circuit for the displayed polynomial.

Lemma 8.5. *Let $n \in \mathbb{Z}$ with $R(n) \neq 0$. If $x \in \mathbb{Q}$ satisfies*

$$\mathrm{Supp}(\alpha_i x + \beta_i, m_i(n)P) \qquad (1 \le i \le 4), \tag{8.9}$$

then $x \notin \mathcal{C}$ and

$$h(x) \le H|R(n)|^c. \tag{8.10}$$

Proof. Each value $\alpha_i x + \beta_i$ is nonzero by the definition of Supp, so the finite point x is not marked. Fix a prime $q \notin S$ not dividing $R(n)$. Proposition 8.2 implies that every valuation $v_q(\alpha_i x + \beta_i)$ is even. Put $\mu_q = \min\{v_q(x), 0\}$. If $v_q(x) \ge 0$, then $\mu_q = 0$. If $v_q(x) < 0$, each α_i is a q-adic unit and each β_i is integral, so

$$v_q(\alpha_i x + \beta_i) = v_q(x).$$

Thus μ_q is even in either case. The four finite normalized contact orders

$$v_q(\alpha_i x + \beta_i) - \mu_q$$

and the contact order $-\mu_q$ at infinity are all even. Consequently every prime in $M_S(x)$ divides $R(n)$, and

$$M_S(x) \mid \prod_{q \mid R(n)} q, \qquad M_S(x) \le |R(n)|.$$

The height estimate (2.7) proves (8.10). □

Let

$$\mathcal{E} = \{n \in \mathbb{Z} : R(n) = 0\} = \mathcal{C} \cap \mathbb{Z}.$$

This is a fixed finite set, possibly empty.

Definition 8.6. For a rational number x and a point code T, require $T \in G$. Define $\mathrm{Eval}(x,T)$ to hold if either

$$\bigvee_{e \in \mathcal{E}} (x = e \ \wedge \ T = eP)$$

holds, or there exist $M_1, \dots, M_4, K \in G$ satisfying

$$\begin{aligned} &M_i = [\alpha_i]T + [\beta_i]P && (1 \le i \le 4),\\ &\mathrm{Supp}(\alpha_i x + \beta_i, M_i) && (1 \le i \le 4),\\ &\Theta_{Y-\kappa(X)}(T,K), \qquad \mathrm{Cong}_3(x,T,K). \end{aligned} \tag{8.11}$$

The empty exceptional disjunction is false.

All multipliers in the first line of (8.11) are fixed integers. The quantifiers range over rational point codes and rational witnesses for the stated formulas.

Proof of Theorem 1.2(ii). Let $x = n \in \mathbb{Z}$ and $T = nP$. If $n \in \mathcal{E}$, use the exceptional branch. Otherwise choose $M_i = m_i(n)P$ and $K = \kappa(n)P$. Proposition 8.2 supplies all four support conditions, the polynomial relation gives the required precision point, and Proposition 8.4 applies because $x - n = 0$. Thus every integer has witnesses with its prescribed point nP.

Conversely, fix $T = nP$ and suppose $\mathrm{Eval}(x,T)$ holds. In the exceptional branch, $T = eP$ implies $n = e$ because P is nontorsion, and hence $x = n$. In the other branch the point identities

force $M_i = m_i(n)P$. The support conditions make these points nonidentity, so $R(n) \neq 0$. Lemma 8.5 yields $h(x) \leq H|R(n)|^c$, and the last two conditions give

$$v_3(x-n) \geq \kappa(n).$$

If $x \neq n$, the reduced numerator of $x-n$ is divisible by $3^{\kappa(n)}$. Since $|R(n)| \geq 1$ and $h(n) \leq n^2$, we obtain

$$\begin{aligned}\kappa(n)\log 3 &\leq h(x-n)\\ &\leq h(x)+h(n)+\log 2\\ &\leq HR(n)^{2c}+n^2+\log 2\\ &< HR(n)^{2c}+n^2+1=\kappa(n).\end{aligned}$$

This contradicts $\log 3 > 1$ and $\kappa(n) > 0$. Therefore

$$\mathrm{Eval}(x, nP) \quad \Longleftrightarrow \quad x = n.$$

Every relation in Definition 8.6 has a fixed existential definition over $\mathbb{Q}$. In particular, the numbers of point variables, support certificates, and local conditions are independent of n and of the precision $\kappa(n)$. Existentially quantifying T therefore defines precisely $\mathbb{Z} \subseteq \mathbb{Q}$. Expanding the fixed formulas and point charts, replacing inequations by equations with reciprocal witnesses, combining conjunctions by sums of squares and disjunctions by products, and clearing fixed coefficient denominators gives one polynomial with integer coefficients, as asserted. □

9. The logical status of the remaining input

Corollary 9.1. *Let $T_{\mathbb{Z}}$ denote the existence of one integer-coefficient existential polynomial defining the ordinary subset $\mathbb{Z} \subset \mathbb{Q}$, as in Theorem 1.2. Then, without additional arithmetic hypotheses,*

$$\mathrm{P} \iff T_{\mathbb{Z}}.$$

Neither side is established in this account.

Proof. Theorem 1.1, proved in Sections 3–5, supplies the other three inputs. The local, multiplication, and evaluation proofs in Sections 6–8 therefore prove $\mathrm{P} \Rightarrow T_{\mathbb{Z}}$.

Conversely, if D is a polynomial defining $\mathbb{Z}$ as in $T_{\mathbb{Z}}$, take

$$\Phi(t): \quad \exists \mathbf{y} \in \mathbb{Q}^N \; D(t,\mathbf{y}) = 0, \qquad S_\Phi = \varnothing.$$

This positive existential formula accepts every integer. Every accepted rational is an integer and has no negative prime valuation. Thus P holds. The converse is not used as an arithmetic input in the forward proof. □

The remaining global existence statement is exactly

$$\begin{aligned}&\exists \text{ one positive existential formula } \Phi(t) \text{ over } \mathbb{Q} \quad \exists \text{ a finite prime set } S_\Phi \quad \text{such that}\\ &\qquad \forall n \in \mathbb{Z}\ \Phi(n), \qquad \forall t \in \mathbb{Q}\ \forall q \notin S_\Phi:\\ &\qquad \Phi(t) \ \wedge \ v_q(t) < 0 \implies v_q(t) \in 2\mathbb{Z}.\end{aligned} \tag{9.1}$$

The formula and exceptional set precede all inputs and all primes. A formula depending on the integer, a finite-range verification, or a predicate excluding poles only at a restricted set of primes would not prove this statement. Its equivalence with $T_{\mathbb{Z}}$ establishes a logical implication from the target to the existence of some suitable predicate. It does not show that any proof must use the particular predicate or elliptic family examined next.

10. The remaining parity input and a direct point construction

Input P asks for one positive existential predicate which accepts every integer and excludes odd-order poles outside a fixed finite set. The source constructs a predicate with the required local soundness; its unresolved step is integer completeness. The statements in this section are unconditional reductions of that candidate construction. They do not assume that its integer witnesses exist.

10.1. **The point problem in the actual formula.** The source fixes finite sets of nonzero rational multipliers m, m_d. For $b \neq 0$ it introduces rational witnesses t, e and puts

$$b' = mb, \qquad d_b = m_d b, \qquad U = b\frac{1 - d_b t^2}{1 + d_b t^2}, \qquad H' = b'(b' - U^2).$$

It requires $1 + d_b t^2 \neq 0$, $eH' \neq 0$, and a rational point with nonzero x-coordinate of squareclass b' on each of

$$E_e : y^2 = x(x - e)(x + 3e), \qquad E_{eH'} : y^2 = x(x - eH')(x + 3eH'). \tag{10.1}$$

The input $b = 0$ has a separate immediate branch. Nonzero conditions are expressible existentially by inverses, and finite multiplier choices are finite disjunctions, so this is a positive existential formula; see [4, Section 3].

Lemma 10.1. *Outside the primes dividing* 6 *or a numerator or denominator of a fixed multiplier, the displayed formula excludes odd-order poles.*

Proof. Suppose $v_q(b) = a < 0$ is odd and the nonzero branch has witnesses. Then $v_q(b') = v_q(d_b) = a$. If $t \neq 0$, the valuation of $d_b t^2$ is odd and hence nonzero. Thus $1 - d_b t^2$ and $1 + d_b t^2$ have the same valuation. The case $t = 0$ gives the same conclusion directly, namely

$$v_q(U) = a, \qquad v_q(b' - U^2) = 2a, \qquad v_q(H') = 3a.$$

The parameters e and eH' have opposite valuation parity. Write the even-valuation one as $l = q^{2c}L$ with L a unit. The change $x = q^{2c}X$, $y = q^{3c}Y$ gives

$$Y^2 = X(X - L)(X + 3L).$$

For $v_q(X) < 0$ the right side has valuation $3v_q(X)$; for $v_q(X) > 0$ it has valuation $v_q(X)$. Either must be even. A nonzero root point has unit X, since $q \neq 3$. Thus every permissible nonzero x has even valuation, whereas its required squareclass b' has odd valuation. This contradiction proves the assertion. The input $b = 0$ has no pole, and the exceptional prime set remains finite over all multiplier choices. □

When a preparation has $b', H' > 0$ and b' is nonsquare, let β, θ be their positive squarefree representatives. Thus $\beta \neq 1$. The next propositions concern exactly these specified parameters, without assuming that such a preparation supplies points. Multiplication of the twist parameter by a rational square induces an isomorphism preserving x-coordinate squareclasses. Consequently the precise remaining requirement is to find one $e \in \mathbb{Q}^*$ such that E_e and $E_{e\theta}$ both have a point of nonzero x-squareclass β.

In the formula, e need not be integral, positive, or squarefree, and need not satisfy the source's Selmer-matrix conditions. Those conditions are one proposed way of supplying witnesses. The same e must, however, work for both curves.

Proposition 10.2. *Let $\beta > 0$ be a nonsquare rational number, and define*

$$k_\beta(s) = \beta\frac{s^2 - \beta}{s^2 + 3\beta} \quad (s \in \mathbb{Q}), \qquad k_\beta(\infty) = \beta.$$

For $e \in \mathbb{Q}^*$, the curve E_e has a rational point with nonzero x-coordinate in $\beta\mathbb{Q}^{*2}$ if and only if

$$e = z^2 k_\beta(s) \quad \text{for some } z \in \mathbb{Q}^*,\ s \in \mathbb{P}^1(\mathbb{Q}).$$

Every value $k_\beta(s)$ in this statement is nonzero.

Proof. Write $x = \beta z^2$ and set $k = e/z^2$, $v = y/z^3$. The curve equation becomes

$$v^2 = \beta(\beta - k)(\beta + 3k).$$

If $k \neq \beta$, put $s = v/(k - \beta)$. Cancelling $k - \beta$ in the resulting identity gives

$$(s^2 + 3\beta)k = \beta(s^2 - \beta),$$

so $k = k_\beta(s)$. The denominator is positive, and the numerator is nonzero because β is nonsquare. If $k = \beta$, use $s = \infty$.

Conversely, for finite s put $k = k_\beta(s)$ and $v = s(k - \beta)$. Substitution verifies the displayed conic equation, and $(\beta z^2, z^3 v)$ is a point on $E_{z^2 k}$. At $s = \infty$ use $(\beta z^2, 0)$. The other permitted nonzero two-torsion case, $x = -3e$, occurs at $s = 0$. Thus the parametrization includes every allowable torsion case; neither the identity nor $x = 0$ is an allowable witness. □

Corollary 10.3. *For β as in Proposition 10.2 and $\theta \in \mathbb{Q}^*$, the paired point requirement is equivalent to*

$$\theta k_\beta(s) = u^2 k_\beta(t), \qquad s, t \in \mathbb{P}^1(\mathbb{Q}), \quad u \in \mathbb{Q}^*. \tag{10.2}$$

For finite s, t, it is equivalently

$$V^2 = \theta F_\beta(s) F_\beta(t), \qquad V \neq 0, \qquad F_\beta(T) = (T^2 - \beta)(T^2 + 3\beta). \tag{10.3}$$

Proof. Apply Proposition 10.2 to write $e = z_1^2 k_\beta(s)$ and $e\theta = z_2^2 k_\beta(t)$, and put $u = z_2/z_1$. Conversely, given (10.2), take $e = k_\beta(s)$, $z_1 = 1$, and $z_2 = u$ in that proposition. Both twist parameters are nonzero.

For finite parameters, clear the nonzero denominators to obtain

$$u^2(s^2 + 3\beta)(t^2 - \beta) = \theta(s^2 - \beta)(t^2 + 3\beta).$$

Multiplying by $(s^2 + 3\beta)(t^2 - \beta)$ and putting $V = u(s^2 + 3\beta)(t^2 - \beta)$ gives (10.3). The inverse substitution is valid because neither factor vanishes at a rational argument. □

The cases $s = \infty$ or $t = \infty$ are retained in (10.2); an argument using only the affine equation must check them separately. If $\theta = 1$, take $s = t$ and $u = 1$. For nonsquare θ, the same diagonal does not give a rational u.

This is an equivalent reformulation of the precise point requirement, not a proof of its solubility. It gives a concrete alternative to the rank argument without narrowing the allowed rational twist parameters. Its geometry also explains an obstacle to applying a positive-rank theorem directly. The curve

$$C_\beta : w^2 = F_\beta(s)$$

has genus one: its degree-two map to $\mathbb{P}^1$ has four distinct simple branch points, so Riemann–Hurwitz gives $2g - 2 = -4 + 4 = 0$. The same is true after multiplying F_β by any nonzero constant. Equation (10.3) is the quotient of

$$w_1^2 = F_\beta(s), \qquad w_2^2 = \theta F_\beta(t)$$

by the simultaneous sign involution, with invariant coordinate $V = w_1 w_2$. Over an algebraic closure this is a product-of-genus-one-curves presentation of a Kummer surface.

For fixed finite rational s, the fiber is a genus-one curve whose leading coefficient need not be a rational square; its points at infinity need not be rational. Such a curve is a torsor under its Jacobian. A positive-rank statement for the Jacobian does not show that this particular torsor is trivial. Thus a theorem producing simultaneous positive ranks cannot replace the paired-point assertion without an additional argument about these specific coverings. That argument has not been established here.

11. Two obstructions to simple replacements for P

11.1. **The congruent-number CM family.** The elementary pole argument also works on the congruent-number family

$$C_l : y^2 = x(x - l)(x + l).$$

This makes it natural to ask whether a CM converse theorem can replace the source's non-CM point-existence argument. Keeping the same prescribed two-descent class encounters the following exact obstruction.

Proposition 11.1. *Let $\beta > 0$ be a nonsquare rational number. There exist $l \in \mathbb{Q}^*$ and a nonroot rational point on C_l with*

$$[x + l] = [\beta], \qquad [x - l] = 1$$

if and only if

$$\beta \in N_{\mathbb{Q}(\sqrt{2})/\mathbb{Q}}(\mathbb{Q}(\sqrt{2})^*).$$

Proof. Write $x + l = \beta w^2$ and $x - l = z^2$, with $w, z \neq 0$. The elliptic equation then implies $x = \beta a^2$ for some $a \neq 0$. Adding the root-difference equations gives

$$\beta(w^2 - 2a^2) = -z^2.$$

Thus $-1/\beta$ is the norm of $(w + a\sqrt{2})/z$. Since $N(1 + \sqrt{2}) = -1$, the number β is a norm.

Conversely, if β is a norm, so is $-1/\beta$. Hence

$$w^2 - 2a^2 = -1/\beta$$

has a rational point. The line parametrization of a nonsingular conic with a point supplies infinitely many rational points, so choose one with $a, w \neq 0$. Set

$$x = \beta a^2, \qquad l = \beta a^2 - 1, \qquad y = \beta a w.$$

Then $x - l = 1$ and $x + l = \beta w^2$, and the elliptic equation follows. Moreover $l \neq 0$, because $\beta a^2 = 1$ would make β a square. All three root factors are nonzero. □

The necessity part of the preceding proof uses only $\beta \neq 0$, so it also applies to negative rational β. Thus the next obstruction includes finite catalogues containing negative multipliers.

Corollary 11.2. *For any fixed finite set $\mathcal{M} \subset \mathbb{Q}^*$, there are infinitely many positive integers b such that none of the classes $[mb]$, $m \in \mathcal{M}$, is a norm from $\mathbb{Q}(\sqrt{2})$.*

Proof. Choose a prime $q \equiv 3 \pmod 8$ outside every multiplier support, and put $b = q$. Dirichlet's theorem gives infinitely many choices. For every $m \in \mathcal{M}$, $v_q(mb) = 1$. But q is inert in $\mathbb{Q}(\sqrt{2})$, and norms from an unramified quadratic local extension have even valuation. Multiplication by a rational square does not change this obstruction. □

For completeness, the nonroot qualification in Proposition 11.1 must also be checked at the four two-torsion points. Extend the Kummer coordinates as $\delta = ([x + l], [x - l])$, using the usual product of root differences at a vanishing factor. Direct evaluation gives

$$\delta(O) = (1, 1), \qquad \delta((0, 0)) = ([l], [-l]),$$
$$\delta((l, 0)) = ([2l], [2]), \quad \delta((-l, 0)) = ([2], [-2l]).$$

For example the vanishing first coordinate at $x = -l$ is the class of $(-l - 0)(-l - l) = 2l^2$, and the vanishing second coordinate at $x = l$ is the class of $(l - 0)(l + l) = 2l^2$. A class $(\beta, 1)$ can arise here only with $[\beta]$ equal to 1, -1, or 2. For the primes $q \equiv 3 \pmod 8$ chosen in Corollary 11.2, every $[mq]$ has odd q-valuation, so none has those three classes. Thus the root points do not evade the stated finite-multiplier obstruction.

Thus replacing the auxiliary family by C_l while keeping its required Kummer class $(\beta, 1)$ fails for infinitely many integer inputs, regardless of the fixed finite multiplier lists. Changing only the prescribed squareclass at 2 cannot remove the inert odd primes. This does not prohibit different Kummer classes or a different CM construction. For example, 3 is not a norm from $\mathbb{Q}(\sqrt{2})$, but C_3 has the point $(3, 0)$ with x-squareclass 3; the obstruction concerns the specified pair of Kummer coordinates.

11.2. **No generic rational-function witnesses.**

Proposition 11.3. *Fix nonzero rational constants m, m_d. The nonzero branch of the source formula in Section 10.1 has no solution over $\mathbb{Q}(B)$ at the generic input $b = B$. This remains true for a finite disjunction of fixed multiplier choices.*

Proof. Use the discrete valuation at infinity, normalized by $v(B) = -1$. Every nonzero rational constant is a unit, including 2 and 3. Hence $v(d_B) = v(b') = -1$. If $t = 0$, then $U = B$. Otherwise $v(d_B t^2) = -1 + 2v(t)$ is odd and nonzero; therefore $1 - d_B t^2$ and $1 + d_B t^2$ have the same valuation. In both cases,

$$v(U) = -1, \qquad v(b' - U^2) = -2, \qquad v(H') = -3.$$

The parameters e and eH' have opposite valuation parity. Select the one of even valuation and write it as $l = \pi^{2a} L$, where L is a unit and π is a uniformizer. Scaling $x = \pi^{2a} X$, $y = \pi^{3a} Y$ gives

$$Y^2 = X(X - L)(X + 3L).$$

Every nonzero X has even valuation. If $v(X) < 0$, the right side has valuation $3v(X)$; if $v(X) > 0$, its other two factors are units, so the valuation is $v(X)$; and if $v(X) = 0$ there is nothing to prove. Each nonzero right side is a square, so the indicated valuations are even. Root points with nonzero X have unit X and also satisfy the conclusion. Scaling does not change valuation parity of the x-squareclass. The required class b' has odd valuation -1, a contradiction. The same argument applies to each fixed choice in a finite disjunction. □

This excludes a uniform rational identity supplying all witnesses as rational functions of the input. It does not exclude a nonuniform arithmetic construction for integer inputs, which is what integer completeness requires.

12. The exact residual Selmer problem

The following implication isolates the missing arithmetic independently of the details of any particular two-descent.

Proposition 12.1. *Let $A/\mathbb{Q}$ be an elliptic curve with full rational two-torsion. Suppose its four two-torsion points give four distinct Kummer classes and*

$$\dim_{\mathbb{F}_2} \mathrm{Sel}_2(A/\mathbb{Q}) = 3.$$

Let γ be a Selmer class outside their span. Then

$$\gamma \text{ is represented by a rational point} \iff \operatorname{rank} A(\mathbb{Q}) = 1 \iff \mathrm{Sha}(A/\mathbb{Q})[2] = 0.$$

Moreover the only possibilities are

$$(\operatorname{rank} A(\mathbb{Q}), \mathrm{Sha}(A/\mathbb{Q})[2^\infty]) = (1, 0) \quad \text{or} \quad (0, \mathbb{Q}_2/\mathbb{Z}_2).$$

In either case the full 2^∞-Selmer corank is one.

Proof. The Kummer sequence and full rational two-torsion give

$$r + \dim_{\mathbb{F}_2} \mathrm{Sha}(A/\mathbb{Q})[2] = 1, \qquad r = \operatorname{rank} A(\mathbb{Q}). \tag{12.1}$$

If $r = 1$, the Kummer map surjects onto the two-Selmer group, so γ is represented. Conversely, a point representing γ cannot have a class in the torsion image: that image has dimension two and is already supplied by the four distinct two-torsion classes. Therefore $r \geq 1$, and (12.1) gives $r = 1$. The equivalence with vanishing of Sha[2] also follows from that equation.

Write $M = \mathrm{Sha}(A/\mathbb{Q})[2^\infty]$. Its subgroup $M[2]$ is finite by the Kummer sequence. Hence M is cofinitely generated over $\mathbb{Z}_2$. For clarity, its compact Pontryagin dual X has finite $X/2X$. Lift generators of this quotient and let Y be their closed $\mathbb{Z}_2$-span. Then $X = Y + 2^j X$ for every j. The subgroups $2^j X$ tend to zero in the compact topology, because every finite subgroup of M has bounded 2-power exponent. Thus $X = Y$ is finitely generated. The structure theorem over $\mathbb{Z}_2$ and duality give a decomposition

$$\mathrm{Sha}(A/\mathbb{Q})[2^\infty] \simeq (\mathbb{Q}_2/\mathbb{Z}_2)^t \oplus F$$

with F finite. The Cassels–Tate pairing is nondegenerate on the quotient by its maximal divisible subgroup. For an elliptic curve, the rational divisor (O) represents the principal polarization, so its polarization obstruction vanishes and the pairing is alternating; see [19, Section 4, Theorem 5, Corollary 7, and Theorem 8]. Different prime-primary components are orthogonal. Consequently the pairing on this finite 2-primary quotient F is perfect, without assuming that the divisible subgroup or the whole Sha is finite. Thus $\dim_{\mathbb{F}_2} F[2]$ is even. Here is the finite-group argument, which does not presume that the restriction of the pairing to $F[2]$ is nondegenerate. If $F \neq 0$, choose $a \in F$ of maximal order 2^k. Nondegeneracy identifies F with its character group, so some $b \in F$ has $\langle a, b\rangle$ of exact order 2^k. Maximality makes the order of b also 2^k. The alternating pairing shows that $C = \langle a, b\rangle \simeq (\mathbb{Z}/2^k\mathbb{Z})^2$ and that its restricted pairing is nondegenerate. Every element of F can be uniquely adjusted by an element of C to be orthogonal to C, so $F = C \oplus C^\perp$. Inducting gives a direct sum of such pairs and the claimed even dimension. Equation (12.1) becomes

$$r + t + \dim_{\mathbb{F}_2} F[2] = 1.$$

Consequently $F = 0$ and $r + t = 1$, giving the two displayed possibilities. Finally the full Selmer exact sequence gives corank $r + t = 1$. □

12.1. **A sufficient remaining assertion for the candidate formula.** There is no need to restrict the auxiliary twist parameter to the positive squarefree integers merely because one attempted proof does so. An exact sufficient remaining assertion for this particular formula can be written without that restriction:

> There exist finite sets $\mathcal{M}, \mathcal{D} \subset \mathbb{Q}^*$ such that for every $b \in \mathbb{Z} \setminus \{0\}$ there are $m \in \mathcal{M}$, $m_d \in \mathcal{D}$ and rational t, e, z_1, z_2, y_1, y_2 with
>
> $$b' = mb, \quad d_b = m_d b, \quad U = b\frac{1 - d_b t^2}{1 + d_b t^2}, \quad H' = b'(b' - U^2),$$
> $$(1 + d_b t^2) e H' z_1 z_2 \neq 0,$$
> $$y_1^2 = b' z_1^2 (b' z_1^2 - e)(b' z_1^2 + 3e),$$
> $$y_2^2 = b' z_2^2 (b' z_2^2 - eH')(b' z_2^2 + 3eH').$$

This assertion has not been proved. If it holds, the finite-disjunction formula in Section 10.1, with a separate $b = 0$ branch, accepts every integer; Lemma 10.1 supplies its pole exclusion. It therefore proves P and, by Theorem 1.2, the full original target. This is a sufficient route to P, not a claim that the displayed formula is necessary or that renaming its completeness problem is progress.

For a preparation with positive nonsquare b' and positive H', the same question is exactly the projective paired equation of Corollary 10.3, with the two squarefree classes denoted β, θ. The two point witnesses must use one common e.

In the finite-descent route, a computation giving the hypotheses of Proposition 12.1 for both E_n and $E_{n\theta}$ does not yet provide these points. It remains to rule out the divisible Sha alternative for both curves at a common admissible n, or otherwise to realize the two prescribed classes. One suitable choice per integer is enough; two unrelated existence statements about the two twists are not. The full proof of the source's pointwise 2-converse companion has not been certified here, and is not used as an established premise. This is a limitation of the present verification, not a claim that the companion theorem is false or that no such converse can be proved.

13. A failed norm construction, with its exact remaining curve

The paired-surface reduction suggests exploiting the special factor $H' = b'(b' - U^2)$ in the preparation. We tested that suggestion explicitly. In this section write $B = b' > 0$, with B nonsquare, and $H = B(B - U^2) \neq 0$. Put

$$f(s) = \frac{s^2 - B}{s^2 + 3B}, \qquad f(\infty) = 1.$$

The paired-point equation is $Hf(s) = r^2 f(t)$; it is the same equation as the one proved in Section 10.3, before replacing H by its squarefree representative.

The identity

$$H = N_{\mathbb{Q}(\sqrt{B})/\mathbb{Q}}(B + U\sqrt{B})$$

suggests the projective substitution

$$t = \frac{B(s + U)}{Us + B}.$$

Its determinant is $B(B - U^2) = H$, so it is an automorphism of the parameter line. A direct calculation gives, with $D_s = (Us + B)^2$,

$$t^2 - B = \frac{H(s^2 - B)}{D_s}, \qquad t^2 + 3B = \frac{BA(s)}{D_s},$$

where

$$A(s) = (B + 3U^2)s^2 + 8BUs + BU^2 + 3B^2.$$

Away from the displayed affine pole, substitution and cancellation of the nonzero factors give the sufficient equation

$$r^2(s^2+3B) = BA(s). \tag{13.1}$$

Homogenizing gives the same condition at $s=\infty$ or $t=\infty$. Thus no projective parameter is silently excluded. Equation (13.1) is a restriction of the paired surface, not an equivalent replacement for all of its rational points.

This substitution does not rationally parametrize the remaining curve. For $U \neq 0$, exact identities are

$$\operatorname{disc} A = -12B(B-U^2)^2, \qquad \operatorname{Res}(A, s^2+3B) = 64B^2U^2(U^2+3B).$$

For the second identity, reduce $A(s)$ modulo s^2+3B to $8BU(s-U)$ and take the product over its two roots. Under $B>0$, $U\neq 0$, and $H\neq 0$, the two quadratics are separable and have disjoint roots. The smooth completion of

$$w^2 = BA(s)(s^2+3B)$$

therefore has genus one, by Riemann–Hurwitz. If $U=0$, the equation splits as $r^2=B^2$ and $H=B^2$ is already a square. If $U^2=B$, the excluded case $H=0$ occurs. No all-integer construction of a point on the nondegenerate genus-one curve was obtained.

Proposition 13.1 (A counterexample to norm-only sufficiency)**.** *For $B=17$, $U=1$, and $H=272$, the displayed norm identity holds, but there is no $e\in\mathbb{Q}^\times$ such that both E_e and E_{272e} have a nonzero x-coordinate of squareclass* 17*.*

Proof. Here $17(17-1)=272=N(17+\sqrt{17})$. At $q=17$ one has $v_q(B)=v_q(H)=1$. Therefore e and eH have opposite valuation parity. Choose the even-valuation parameter $l=q^{2a}L$ with L a unit, and scale the point coordinates by $x=q^{2a}X$, $y=q^{3a}Y$. On

$$Y^2 = X(X-L)(X+3L)$$

a nonzero X has even valuation. For negative valuation the right side has valuation $3v_q(X)$; for positive valuation its other factors are units, so its valuation is $v_q(X)$; zero valuation is already even. But scaling by q^{2a} preserves the parity of the x-valuation, whereas the required squareclass 17 has odd valuation. This is a contradiction. □

This example fails the preparation's extra mutual-square conditions. It refutes only the assertion that the norm identity alone guarantees the two points. It does not refute the prepared paired-point statement, P, or the main theorem.

14. Restricted contact primes cannot replace the full height estimate

A possible substitute for P would control only primes inert in finitely many fixed quadratic fields. The following independent result rules out using the corresponding restricted odd-contact radical in place of the full radical in H. It does not obstruct the full-prime argument.

Fix any five distinct rational marked points and primitive forms L_i as in H. Let $K_1,\dots,K_r$ be finitely many fixed quadratic fields, let $\mathcal{I}$ be the set of primes inert in at least one of them, and let S be finite. Define $M_{S,\mathcal{I}}(x)$ by counting only primes in $\mathcal{I}\setminus S$ with an odd normalized contact.

Theorem 14.1. *There is a fixed positive integer K and rational points $x\in\mathbb{P}^1(\mathbb{Q})\setminus\mathcal{C}$ of arbitrarily large height such that $M_{S,\mathcal{I}}(x)\mid K$. In particular there is no finite-valued function $B:\mathbb{Z}_{>0}\to\mathbb{R}$ such that*

$$h(x) \le B(M_{S,\mathcal{I}}(x))$$

for all such rational points.

Proof. Choose an integer a different from the finite marked points, and put $k_0=1$, $k_i=L_i(a,1)\neq 0$. Let D be a positive multiple of the absolute discriminants of all K_j. Primes congruent to 1 (mod D) split in each field. Choose a positive integer M divisible by $30D\prod_{i=1}^4|k_i|$ and define the five affine forms

$$\psi_i(U,V) = 1+\frac{M}{k_i}L_i(U,V), \qquad 0\le i\le 4.$$

Their coefficients are integral. Their nonzero linear parts are pairwise nonproportional because the five marked points are distinct. For $p = 2, 3, 5$ the forms are all $1 \pmod p$. For $p > 5$, each zero set is empty or an affine line in $\mathbb{F}_p^2$, so their union has at most $5p < p^2$ points. The system is therefore locally admissible at every prime. For all sufficiently large p the five directions remain distinct, and the local factor is

$$(1 - 1/p)^{-5}(1 - 5/p + O(p^{-2})) = 1 + O(p^{-2}).$$

Its singular product is positive.

At $(U, V) = (a, 1)$, each homogeneous part has the positive value M. Choose a small fixed rectangle $\mathcal{B}$ of positive area around this point on which all these parts lie between $M/2$ and $3M/2$. The finite-complexity theorem for linear prime patterns [14, 15, 16, 17] gives

$$\#\{(U, V) \in X\mathcal{B} \cap \mathbb{Z}^2 : \psi_0(U, V), \dots, \psi_4(U, V) \text{ prime}\} \gg \frac{X^2}{(\log X)^5}.$$

All coefficients and the rectangle are fixed before X tends to infinity. For completeness, the von Mangoldt asymptotic yields this ordinary prime count because proper prime powers have $O(\sqrt{X} \log X)$ possible values, each with $O(X)$ lattice preimages; their weighted contribution is $o(X^2)$. The hypotheses just checked are essential: no one-variable prime-tuples conjecture is being invoked.

The equation $\psi_0 = \psi_1$ describes a genuine affine line. Remove its $O(X)$ lattice points. For a remaining prime tuple put

$$u = a + MU, \qquad v = 1 + MV.$$

Then $L_i(u, v) = k_i\psi_i(U, V)$ for all five indices, and $v = \psi_0$ is a positive prime. For large X it exceeds every $|k_i|$. If $v \mid u$, it would divide $L_1(u, v) = k_1\psi_1$; this is impossible because $v \nmid k_1$ and ψ_1 is a different prime. Thus (u, v) is primitive and

$$h(u/v) \geq \log v \longrightarrow \infty.$$

All five contact values are nonzero. Every variable prime ψ_i is $1 \pmod D$ and hence splits in every K_j. The remaining contact primes divide the fixed product $\prod_{i=1}^4 k_i$. Consequently

$$M_{S,\mathcal{I}}(u/v) \mid K := \operatorname{rad}\left(\prod_{i=1}^4 |k_i|\right).$$

A finite-valued function has a finite maximum on the finitely many divisors of K, contradicting the unbounded heights. This proves both assertions. □

14.1. **An explicit checked instance.** Take $\mathcal{C} = \{\infty, 0, 1, -1, 2\}$, $K_1 = \mathbb{Q}(\sqrt{2})$, and $S = \{2, 3\}$. With $a = 3$, $D = 8$, $(k_0, \dots, k_4) = (1, 3, 2, 4, 1)$ and $M = 5760$, the pair $(U, V) = (26, 11)$ gives

$$x = \frac{149763}{63361}.$$

Its five primitive contact values are

$$63361, \quad 3 \cdot 49921, \quad 2 \cdot 43201, \quad 4 \cdot 53281, \quad 23041.$$

All five displayed prime factors are prime and $1 \pmod 8$. The restricted radical is therefore 1. The accompanying script verifies primality by trial division through the integer square root, together with the exact identities and coprimality. This finite example illustrates the theorem; it is not the proof of unboundedness.

14.2. **Why an elliptic family does not replace the height family.**

Proposition 14.2. *Let Y be a connected smooth projective curve over a number field and $\mathcal{E} \to Y$ an elliptic scheme. Its j-invariant is constant on each geometric component. If $\pi : Y \to \mathbb{P}^1$ is nonconstant, there is no constant C for which*

$$h(\pi(y)) \leq C(1 + \max\{0, h_F(\mathcal{E}_y)\}) \qquad (y \in Y(\overline{\mathbb{Q}})).$$

Proof. The j-invariant is a regular function on a proper curve and hence constant on each geometric component. Elliptic curves with the same j-invariant are geometrically isomorphic, so stable Faltings height is fixed there. A nonconstant map from a projective curve to $\mathbb{P}^1$ is surjective. Lift rational base points of unbounded height to algebraic points on a component where the map is nonconstant. The proposed inequality fails. □

The Legendre family exhibits the corresponding local problem. Its invariant is

$$j(t) = 256\frac{(1-t+t^2)^3}{t^2(1-t)^2}.$$

At an odd prime q, take $t = q^{2n}$ with $n \geq 1$. The contact with 0 is even, but $v_q(j(t)) = -4n$. The curve cannot acquire good reduction in any finite extension, because its negative j-valuation remains negative. Thus an even contact does not remove its multiplicative degeneration. These arguments explain why the complete abelian-surface family in Section 4 has a substantive role. They do not exclude all other uses of elliptic curves.

15. Final mathematical status

The unconditional arithmetic results are Proposition 3.1, Theorem 4.1, and Proposition 5.1. They have the original input scopes, with H specialized to the proved exponent 24 and R applied to the same five marked points. Their proofs do not use P, an integer definition, or the unverified pointwise converse. They therefore establish Theorem 1.1 without circularity.

The full original conclusion is Theorem 1.2, with precisely P remaining. The proof includes the complete local certificates, the multiplication formula, support completeness, the fixed local ring, variable precision, and the exceptional marked-integer branches. It does not rely on a finite computation for an infinite assertion.

Equation (9.1) is the exact unresolved proposition. Section 12.1 gives one sufficient concrete global point formulation for the proposed predicate; it is not an additional proved theorem. The norm example and the restricted-height theorem refute specified substitute assertions only. No claim of a counterexample to the original theorem, of impossibility, or of an unconditional solution follows.

The supplied verification program checks the displayed finite identities and examples with exact arithmetic. The sieve, height bound, and rank calculation are proved above independently of those checks. The separate `verification_note.md` records source versions, repairs, computation scope, and production checks. No unconditional replacement proof is supplied, because the remaining global assertion has not been established.

Appendix A. Formal parameters and the estimates used in the proof

The distinction between two local arguments is useful. For completeness of the support relation, a point need only approach the identity at finitely many places; this works even at bad reduction. For the comparisons used in multiplication and at 3, we use the stronger valuation estimates available at odd unramified places.

Lemma A.1 (Local neighborhoods of the identity)**.** *For the displayed equation of E, at every finite place w there are arbitrarily small open subgroups of $E(F_w)$ parametrized by sufficiently small values of $z = -x/y$. In this parameter the group law has integral coefficients, and for every signed integer a,*

$$[a](Z) = aZ + Z^2F_a(Z), \qquad F_a(Z) \in \mathcal{O}_{F_w}[[Z]]. \tag{A.1}$$

The function $1/x$ extends to the identity with value zero, and $z([a]T)/z(T)$ extends there with value a. Each of the indicated open subgroups has finite index in $E(F_w)$.

At a place of good reduction, z parametrizes the whole kernel of reduction by the maximal ideal. For a nonidentity point with $x \neq 0$, membership in this kernel is equivalent to $w(1/x) > 0$.

Proof. The Weierstrass coefficients are integral at every finite place. The usual formal construction at the smooth point O therefore gives an integral group law, whether or not the rest of the

reduced cubic is smooth [5, Chapters IV and VII]. One can see the parameter expansion directly by putting $w_0 = -1/y$:

$$x = z/w_0, \qquad w_0 = z^3 - d^2 z w_0^2.$$

The second equation has a unique solution $w_0 \in z^3\mathcal{O}_{F_w}[[z]]$ with leading term z^3, found recursively. Hence $1/x = w_0/z = z^2 + \cdots$ extends with value zero. The integral addition and inverse series give (A.1), including negative a, and show that sufficiently small parameter balls are subgroups. Their power series converge there. Dividing (A.1) by Z proves the stated extension of the quotient.

These parameter balls are open neighborhoods of O in the analytic group $E(F_w)$. This group is compact because E is projective. An open subgroup of a compact group has finite index: its cosets form an open cover, and a finite subcover contains all cosets. At good reduction the formal group identifies the full kernel with the maximal ideal. The expansions give $w(x) = -2w(z) < 0$ there. Conversely, a pole of x reduces to the unique point at infinity on the smooth Weierstrass model. This proves the last assertion. □

Lemma A.2 (Valuation preservation and the logarithmic unit)**.** *Let q be odd and let w be an unramified place above q, normalized by $w(q) = 1$. Suppose the curve has an integral equation at w. For Z in the maximal ideal, the formal logarithm converges; for $Z \neq 0$,*

$$\ell(Z)/Z \in 1 + q\mathcal{O}_{F_w}, \qquad w(\ell(Z)) = w(Z). \tag{A.2}$$

On the formal group the logarithm is a homomorphism. These assertions hold also for the conjugate curve.

Proof. Write the invariant differential in the formal parameter as $(1 + \sum_{j\geq 1} b_j Z^j)\, dZ$, with b_j integral. Integrating gives

$$\ell(Z) = Z + \sum_{t\geq 2} c_t Z^t, \qquad w(c_t) \geq -v_q(t).$$

The value group is $\mathbb{Z}$. For $w(Z) \geq 1$ and $t \geq 2$,

$$w(c_t Z^{t-1}) \geq (t-1)w(Z) - v_q(t) \geq 1.$$

The bounds tend to infinity; the last inequality follows from $t - 1 - v_q(t) \geq 1$ for odd q. This proves convergence and (A.2). The formal identity defining the logarithm is an identity of convergent series on this group and therefore makes it a homomorphism. Conjugation changes none of the estimates. □

Lemma A.3 (Ratios of parameters and truncated logarithms)**.** *Let q be odd and unramified in F, and suppose the displayed curve has good reduction at $w \mid q$. Normalize $w(q) = 1$. If $U = aP$ and $V = bP$ are formal at w, where $a, b \in \mathbb{Z}$ and $b \neq 0$, then*

$$w\left(\frac{z(U)}{z(V)}\right) = v_q(a/b). \tag{A.3}$$

For every integer $k \geq 1$ with $q > k + 1$, if $v_q(a/b) \geq 0$, then

$$\ell_k(z(V)) \neq 0, \qquad w\left(\frac{\ell_k(z(U))}{\ell_k(z(V))} - \frac{a}{b}\right) \geq k. \tag{A.4}$$

The same statements hold on E^σ with P^σ.

Proof. The group identity $[b]U = [a]V$ and Lemma A.2 give $b\ell(z(U)) = a\ell(z(V))$. Since $b \neq 0$ and P is nontorsion, $V \neq O$ and $\ell(z(V)) \neq 0$. Taking valuations proves (A.3), with the infinite-valuation convention when $a = 0$.

For $t > k$ and $q > k + 1$,

$$t - 1 - v_q(t) \geq k.$$

Indeed this is immediate when $v_q(t) = 0$; if $j = v_q(t) \geq 1$, then $t - 1 - j \geq q^j - 1 - j \geq q - 2 \geq k$. The coefficient bound in the preceding proof consequently gives

$$w(\ell(Z) - \ell_k(Z)) \geq w(Z) + k \qquad (w(Z) \geq 1). \tag{A.5}$$

In particular $w(\ell_k(z(V))) = w(z(V))$, proving nonvanishing. Put $\lambda = a/b$. If $a \neq 0$ and $v_q(\lambda) \geq 0$, then $w(z(U)) \geq w(z(V))$ by (A.3). Subtract the two logarithmic tails in $\ell(z(U)) = \lambda\ell(z(V))$ and divide by $\ell_k(z(V))$. Equation (A.5) gives the claimed error at least k. If $a = 0$ the comparison is exact. The conjugate proof is identical. □

The coefficients of ℓ_k are effective elements of F. The recursion for w_0 above determines the invariant differential $dx/(2y)$ to any prescribed order, and termwise integration determines ℓ_k. Thus the fixed polynomial ℓ_c used in the main text requires finitely many rational operations. The integer k in the truncation lemma is positive; the separate congruence precision in Section 8 may equal zero.

Appendix B. Explicit rational encoding of the field and elliptic points

We record finite existential encodings for the algebraic operations used in the proof. Every variable in these encodings ranges over $\mathbb{Q}$; an F-valued variable or a point variable abbreviates a fixed tuple of rational variables.

B.1. **The quadratic field and its rational projection.** Represent $u = u_0 + u_1\sqrt{2} \in F$ by $(u_0, u_1) \in \mathbb{Q}^2$. Then

$$\begin{aligned}(u_0, u_1) + (v_0, v_1) &= (u_0 + v_0, u_1 + v_1),\\ (u_0, u_1)(v_0, v_1) &= (u_0v_0 + 2u_1v_1, u_0v_1 + u_1v_0),\\ (u_0, u_1)^\sigma &= (u_0, -u_1).\end{aligned}$$

Thus an F-equation is a pair of rational equations. The rational projection used in the main argument is

$$\rho(u_0 + u_1\sqrt{2}) = u_0 = \frac{1}{2}\,\mathrm{Tr}_{F/\mathbb{Q}}(u).$$

The field norm is $\mathrm{N}_{F/\mathbb{Q}}(u) = u_0^2 - 2u_1^2$, which vanishes exactly when $u = 0$. Consequently an F-inequation $u \neq 0$ is expressed by $\mathrm{N}_{F/\mathbb{Q}}(u)s = 1$ for an auxiliary $s \in \mathbb{Q}$. The graph of inversion is given by the two rational equations expressing $uv = 1$. These interpretations do not require a rational square root of 2.

B.2. **Canonical point codes.** A code for a point of E consists of a rational flag ϵ and two F-coordinates x, y, hence five rational components. Its validity is the finite condition

$$\mathrm{Pt}_E(\epsilon, x, y): \quad (\epsilon = 0 \ \wedge\ x = y = 0) \ \vee\ (\epsilon = 1 \ \wedge\ y^2 = x^3 - d^2x).$$

The first branch is the unique code for O; the second is the unique affine code for the finite point with coordinates (x, y). In particular, the finite two-torsion point $(0, 0)$ has flag 1 and is distinct from the identity code.

Equality of valid codes is equality of their five rational components. For codes with components $q_1, \ldots, q_5$ and $q_1', \ldots, q_5'$, inequality is equivalent to

$$\sum_{i=1}^{5}(q_i - q_i')^2 \neq 0.$$

A reciprocal witness makes this an existential polynomial condition. Conjugation acts componentwise by the displayed rational linear map and gives the corresponding code on E^σ, whose equation has d replaced by d^σ.

B.3. **Addition, inverses, and fixed integer multiples.** For valid codes U, V, W, the equation $W = U + V$ is expressed by a finite disjunction of the following cases. If $U = O$, impose $W = V$; if $V = O$, impose $W = U$. In every remaining branch require U and V to be finite. If $x_U = x_V$ and $y_U + y_V = 0$, impose $W = O$; this includes doubling a two-torsion point.

If $x_U \neq x_V$, require W to be finite and introduce $\lambda \in F$ satisfying

$$\begin{aligned}(x_V - x_U)\lambda &= y_V - y_U,\\ x_W &= \lambda^2 - x_U - x_V,\\ y_W &= \lambda(x_U - x_W) - y_U.\end{aligned}$$

If $x_U = x_V$, $y_U = y_V$, and $y_U \neq 0$, require W to be finite and use

$$2y_U\lambda = 3x_U^2 - d^2,$$
$$x_W = \lambda^2 - 2x_U,$$
$$y_W = \lambda(x_U - x_W) - y_U.$$

These cases exhaust the group law on the nonsingular curve: finite points with the same x-coordinate have equal or opposite y-coordinates. All the displayed equations and inequations have the rational encodings already described. These are the usual addition formulas [5, Chapter III, §2].

Inversion fixes O and sends (x, y) to $(x, -y)$ on the finite chart. For any fixed integer a, a fixed finite sequence of additions and, if needed, an inversion defines the graph of $[a]$. Thus membership in the chosen group $G = m_0E(F)$ is expressed by

$$\mathrm{Pt}_E(T) \ \wedge \ \exists U\big(\mathrm{Pt}_E(U) \ \wedge \ [m_0]U = T\big),$$

where m_0 is the fixed integer from Input E. No negation of this existential membership condition is used.

B.4. **Rational functions and their domains.** On the identity branch define $z(O) = 0$. On a finite branch with $y \neq 0$, the equation $yz = -x$ defines $z = -x/y$. On a finite branch with $x \neq 0$, the equation $xt = 1$ defines $t = 1/x$. A quotient A/B is represented by a variable D and the conditions $B \neq 0$, $BD = A$, over F or $\mathbb{Q}$ as appropriate. The projection ρ then selects the first rational component of its F-argument.

Equations for a rational function remain inside the same disjunct as their domain restrictions. In particular, reciprocal-coordinate equations are not imposed on identity branches. All points to which $1/x$ or a nonzero z-denominator is applied in the main argument belong to $G \setminus \{O\}$ or $G^\sigma \setminus \{O\}$, on the corresponding curve. Their coordinates x, y are nonzero, since points with $x = 0$ or $y = 0$ on either displayed curve are two-torsion and both groups are torsion-free. The witnesses used to encode field arithmetic, point operations, and inverses are not subjected to pole-parity conditions unless a predicate explicitly lists them. Each abbreviation therefore expands into a finite existential combination of rational polynomial equations and inequations.

Appendix C. An explicit definition of the fixed local ring

The congruence construction needs only one fixed local ring. The following formula is a direct specialization of classical local–global theory for quadratic forms. We give its proof, including the local obstruction at 3.

Lemma C.1. *For every $t \in \mathbb{Q}$,*

$$t \in \mathbb{Z}_{(3)} \quad \Longleftrightarrow \quad \exists u, v, w \in \mathbb{Q} : \quad u^2 + v^2 + 3w^2 = 1 + 6t^2,$$

where $\mathbb{Z}_{(3)} = \{t \in \mathbb{Q} : v_3(t) \geq 0\}$.

Proof. Put $q(U, V, W) = U^2 + V^2 + 3W^2$. First consider its values over $\mathbb{Q}_3$. If u, v are not both zero and $a = \min\{v_3(u), v_3(v)\}$, then

$$v_3(u^2 + v^2) = 2a.$$

Indeed, after division by 3^{2a}, the two squares reduce to 0 or 1 modulo 3, and at least one reduces to 1; their sum cannot vanish. The valuation of $3w^2$, when finite, is odd. Thus the two summands $u^2 + v^2$ and $3w^2$ have distinct valuations whenever both are nonzero. It follows that whenever $q(u, v, w)$ has odd valuation e, its normalized unit $3^{-e}q(u, v, w)$ is congruent to 1 modulo 3. The same conclusion holds when $u = v = 0$.

If $v_3(t) = -h < 0$, write $t = 3^{-h}\eta$ with $\eta \in \mathbb{Z}_3^\times$ and $h \geq 1$. Then

$$1 + 6t^2 = 3^{1-2h}\big(3^{2h-1} + 2\eta^2\big).$$

Its valuation is the odd integer $1 - 2h$, and its normalized unit is 2 modulo 3. It cannot be represented by q over $\mathbb{Q}_3$, and hence cannot be represented over $\mathbb{Q}$. Thus rational witnesses imply $t \in \mathbb{Z}_{(3)}$.

Conversely, suppose $v_3(t) \geq 0$, and put $A = 1 + 6t^2 > 0$. We check that q represents A over every completion of $\mathbb{Q}$. At 3, the number A is congruent to 1 modulo 3, so Hensel's lemma [20, Theorem 2.1] gives $s \in \mathbb{Z}_3$ with $s^2 = A$; take $(u, v, w) = (s, 0, 0)$.

For an odd prime $p \neq 3$, the two subsets

$$\{a^2 : a \in \mathbb{F}_p\}, \qquad \{-3 - b^2 : b \in \mathbb{F}_p\}$$

each have $(p+1)/2$ elements, so they intersect. We obtain a nonzero solution of $q(U, V, W) = 0$ modulo p with $W = 1$. At least one of U, V is nonzero, because $p \neq 3$; fixing the other two coordinates and applying Hensel's lemma lifts this solution to $\mathbb{Q}_p$. Thus q is isotropic over $\mathbb{Q}_p$. At 2, the unit $-7 \equiv 1 \pmod 8$ is a square in $\mathbb{Q}_2$, so $(\sqrt{-7}, 2, 1)$ is a nonzero isotropic vector [20, Theorem 4.5].

A nondegenerate isotropic quadratic form represents every scalar: if $q(\mathbf{v}) = 0$ with $\mathbf{v} \neq 0$, nondegeneracy supplies $\mathbf{w}$ with $B(\mathbf{v}, \mathbf{w}) \neq 0$, where B is its associated symmetric bilinear form. The expression $q(\mathbf{w} + \lambda \mathbf{v}) = q(\mathbf{w}) + 2\lambda B(\mathbf{v}, \mathbf{w})$ then takes every value as λ varies. Applying this observation to q proves local representability of A at all finite primes other than 3. Over $\mathbb{R}$, representability follows from $A > 0$.

The homogeneous quadratic form

$$U^2 + V^2 + 3W^2 - AS^2$$

therefore has a nonzero zero over every completion of $\mathbb{Q}$. By the Hasse–Minkowski theorem it has a nonzero zero over $\mathbb{Q}$ [21, Theorem 11.12]. At such a rational zero, S cannot vanish: the form $U^2 + V^2 + 3W^2$ is positive definite over $\mathbb{R}$. Dividing by S^2 gives the required rational witnesses. □

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